

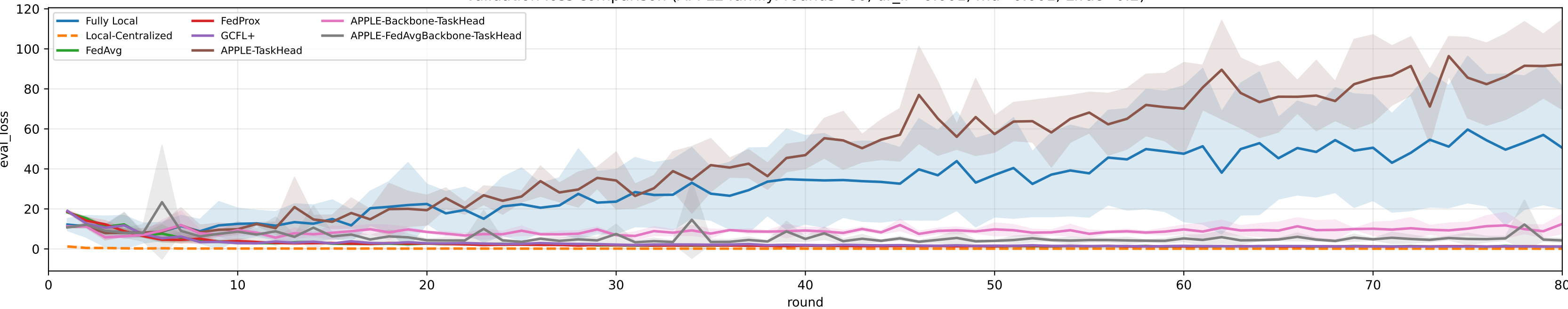
Controlled q-label heterogeneity: High | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

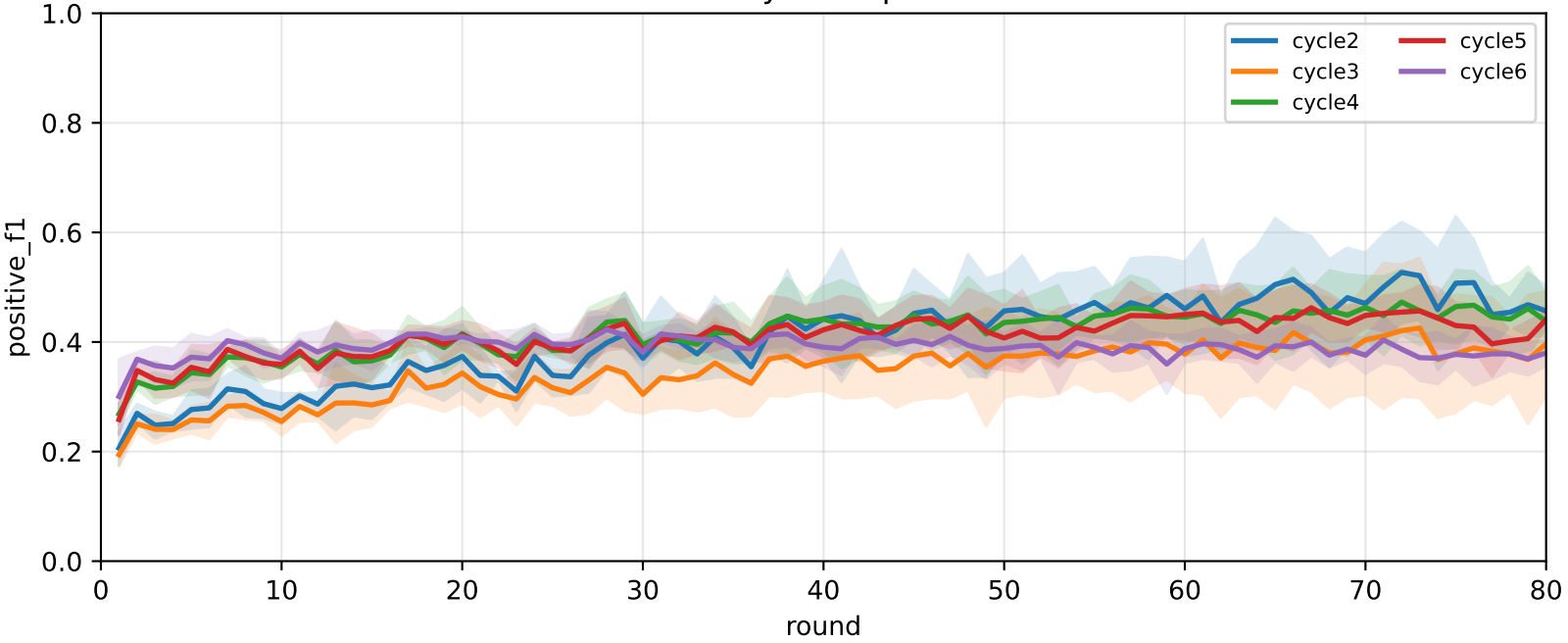
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.6% ± 1.8%	27.6% ± 1.8%	22.4% ± 1.5%	27.5% ± 2.1%	27.3% ± 1.2%	26.7% ± 1.2%	30.2% ± 1.5%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	<u>59.3% ± 1.9%</u>	<u>59.9% ± 1.9%</u>	60.2% ± 2.0%	<u>64.9% ± 5.2%</u>	<u>61.3% ± 2.4%</u>	59.8% ± 2.6%	<u>53.2% ± 3.1%</u>
FedProx	57.5% ± 1.1%	58.3% ± 0.7%	<u>61.7% ± 2.8%</u>	60.2% ± 4.0%	60.2% ± 1.4%	58.5% ± 1.9%	50.8% ± 2.3%
GCFL+	57.4% ± 2.5%	57.9% ± 2.7%	55.8% ± 1.9%	60.8% ± 5.9%	60.2% ± 1.4%	<u>61.7% ± 4.5%</u>	51.1% ± 0.8%
APPLE-TaskHead	31.1% ± 2.6%	31.1% ± 2.6%	25.6% ± 1.9%	31.0% ± 3.0%	31.4% ± 3.3%	30.7% ± 2.5%	34.1% ± 2.9%
APPLE-Backbone-TaskHead	36.0% ± 6.5%	36.0% ± 6.5%	30.7% ± 7.3%	36.7% ± 7.2%	36.5% ± 6.4%	35.9% ± 5.6%	39.7% ± 5.5%
APPLE-FedAvgBackbone-TaskHead	57.2% ± 5.9%	58.4% ± 6.6%	61.4% ± 11.5%	59.8% ± 8.3%	56.5% ± 4.6%	59.3% ± 5.6%	50.2% ± 2.2%

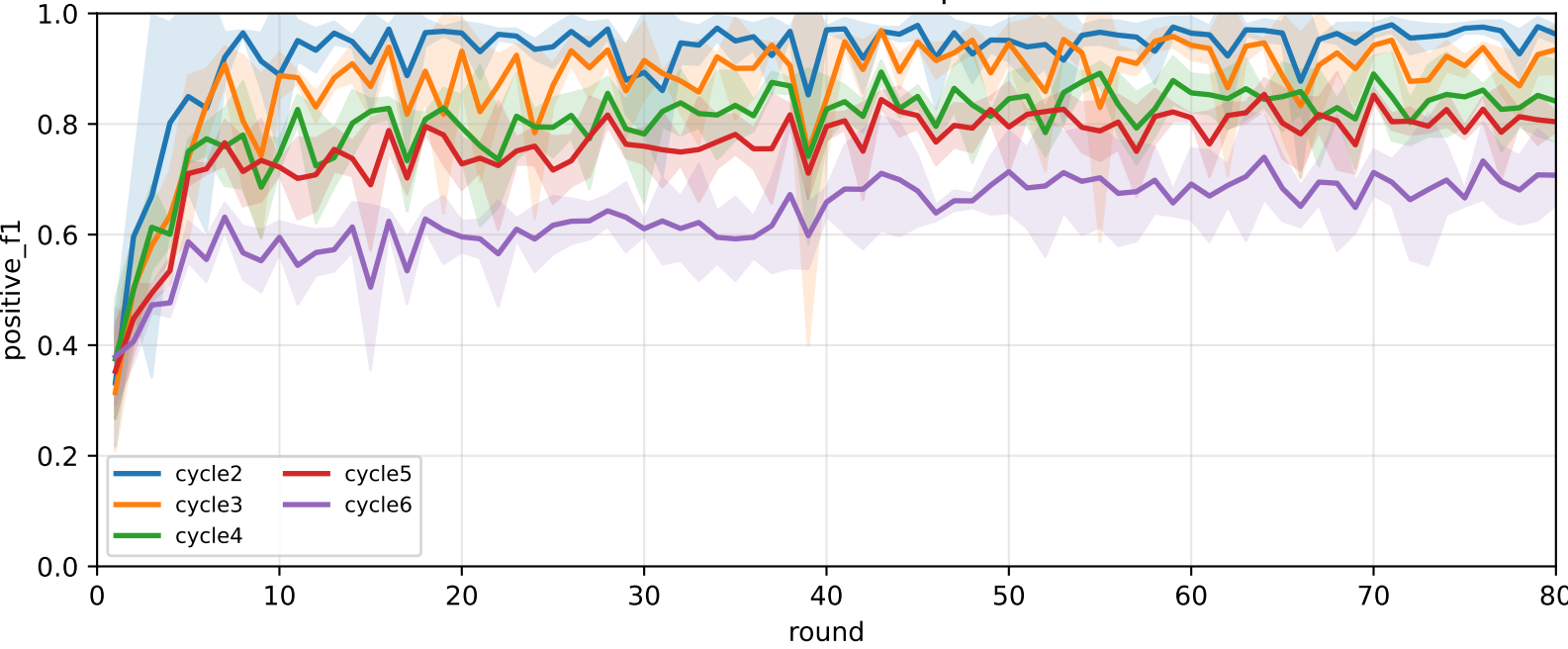
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



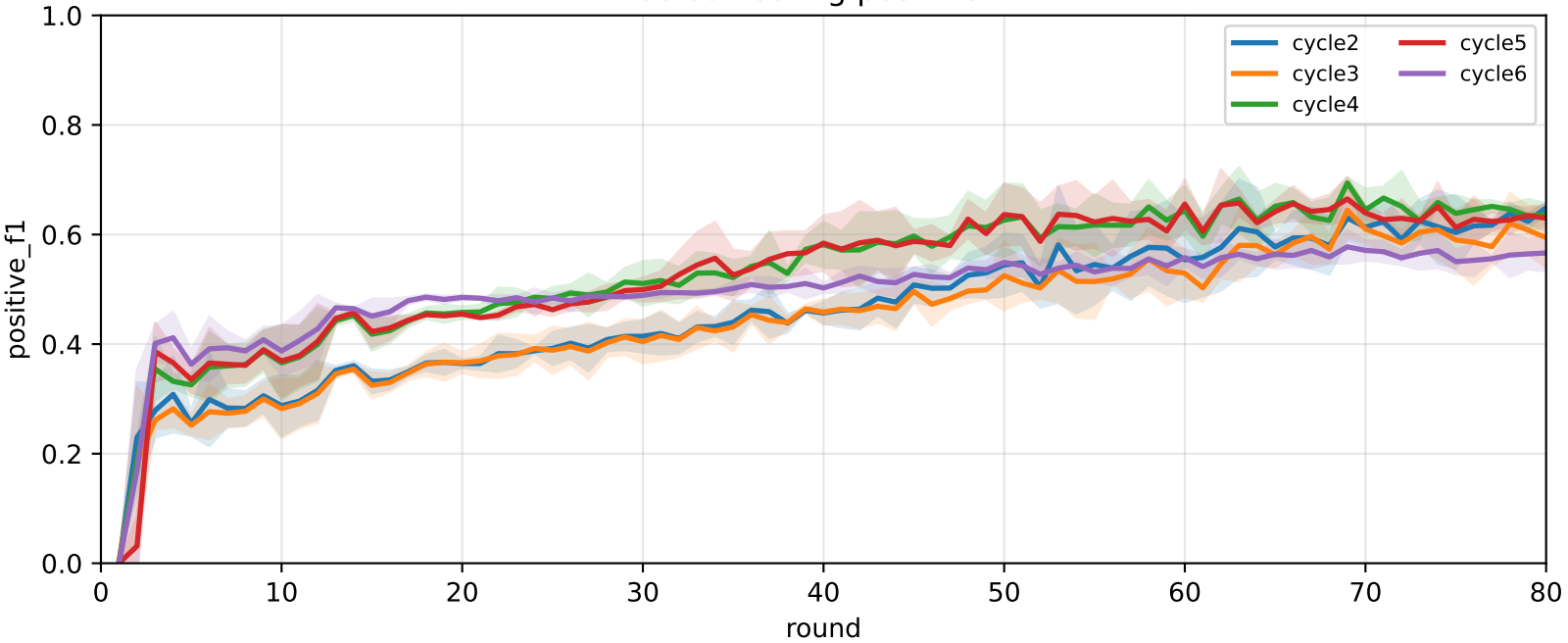
Pooled Fully Local positive F1



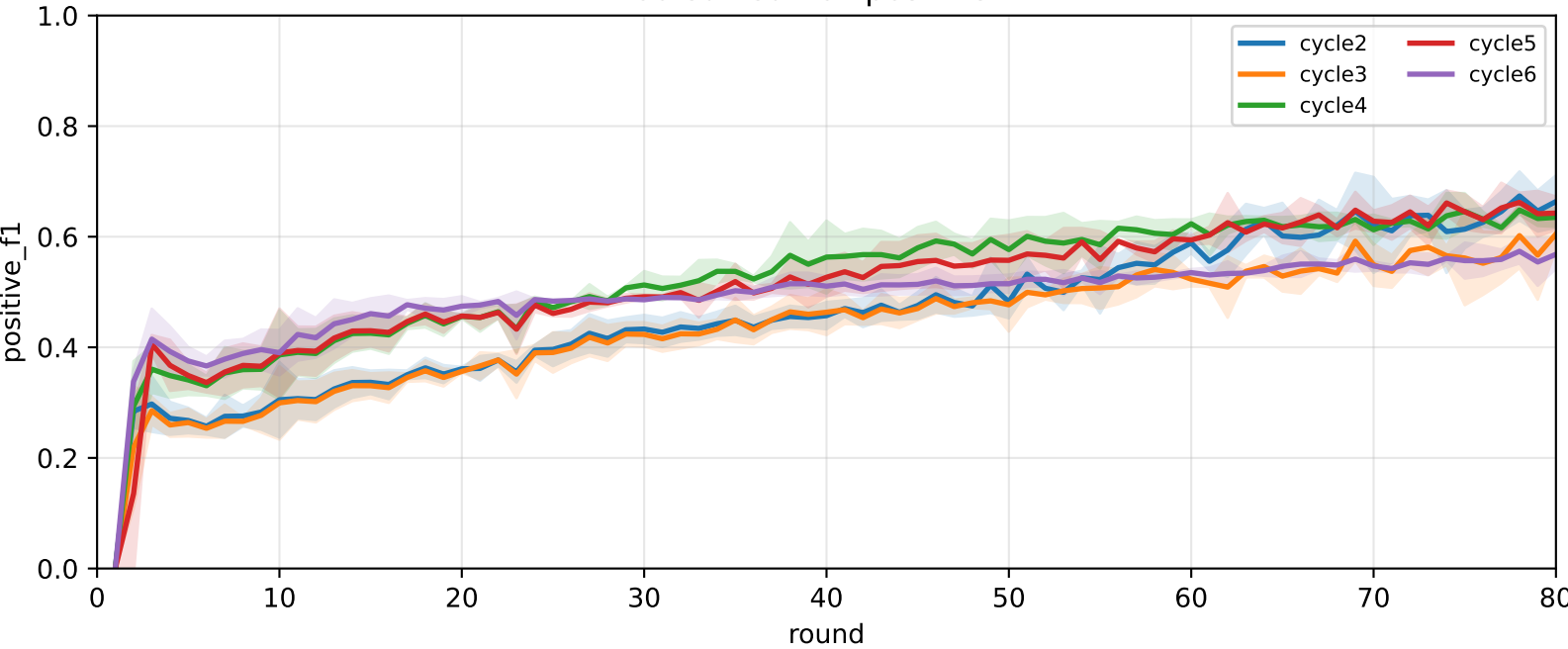
Pooled Local-Centralized positive F1



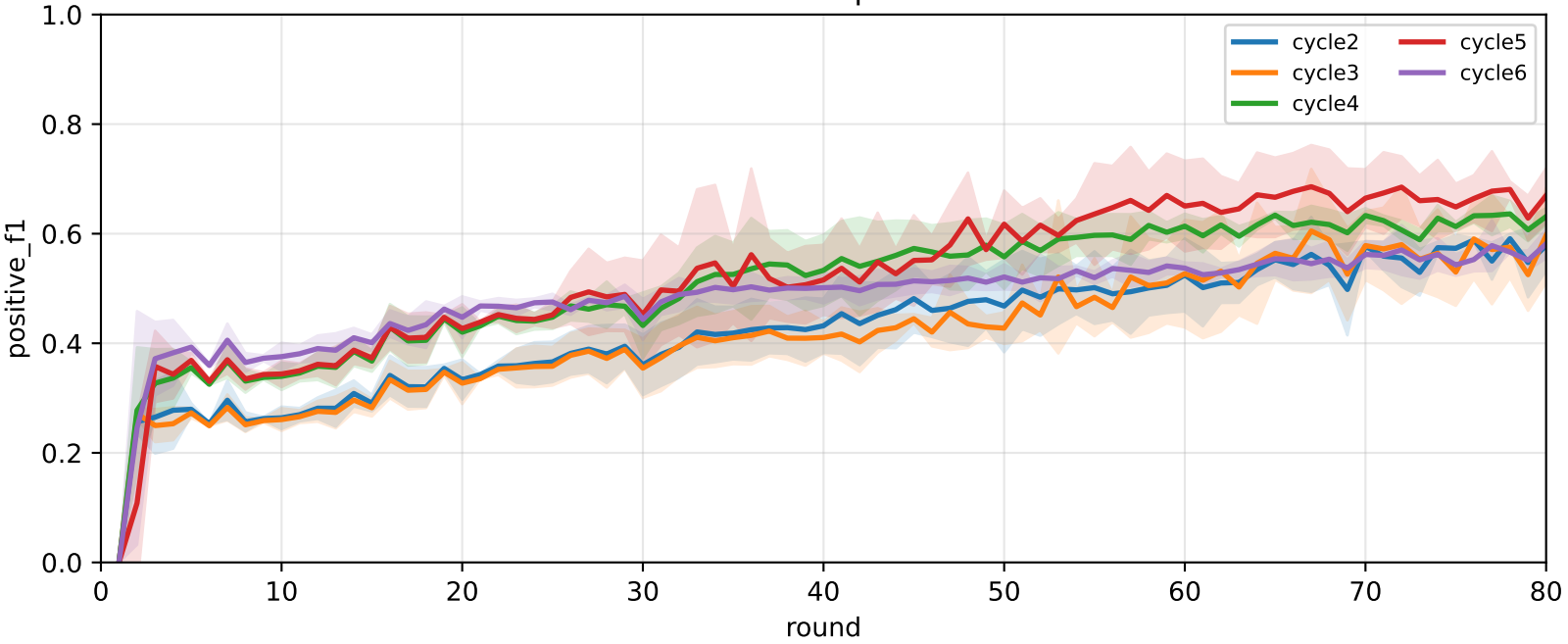
Pooled FedAvg positive F1



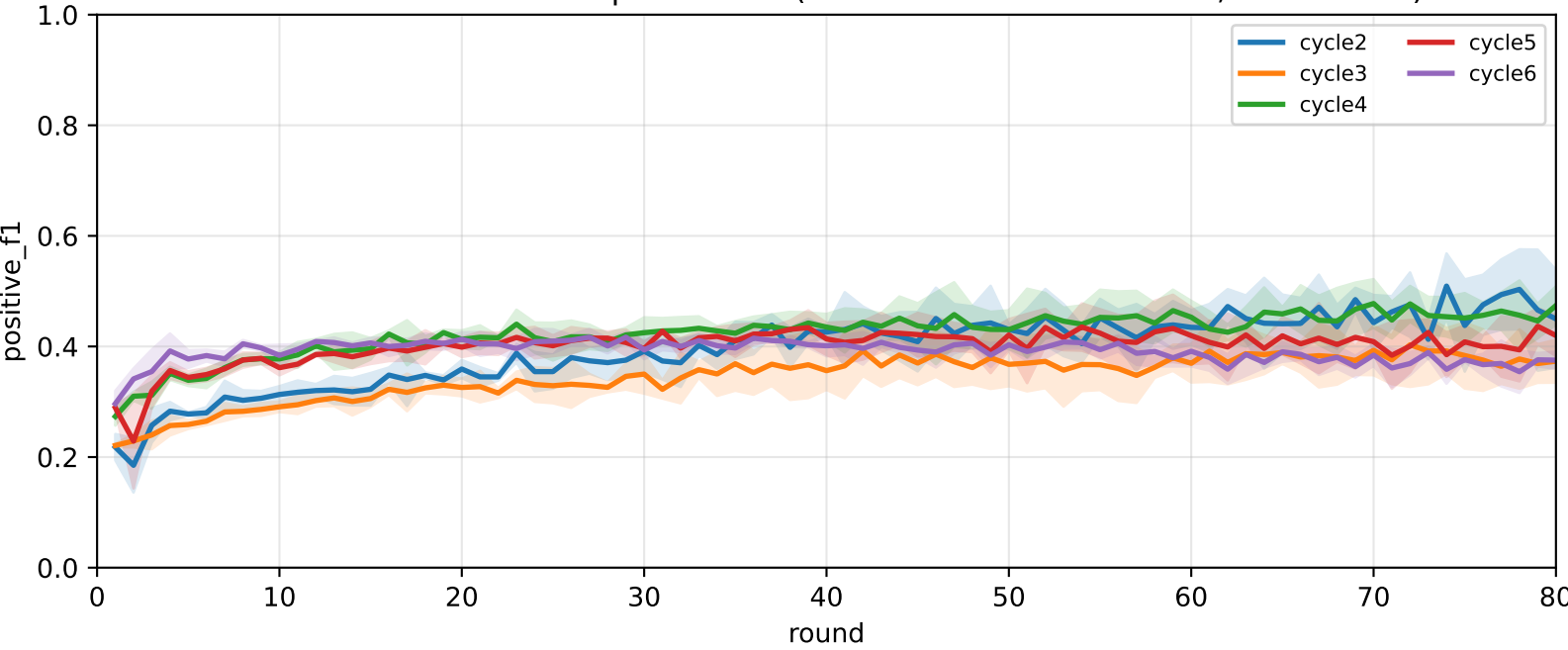
Pooled FedProx positive F1



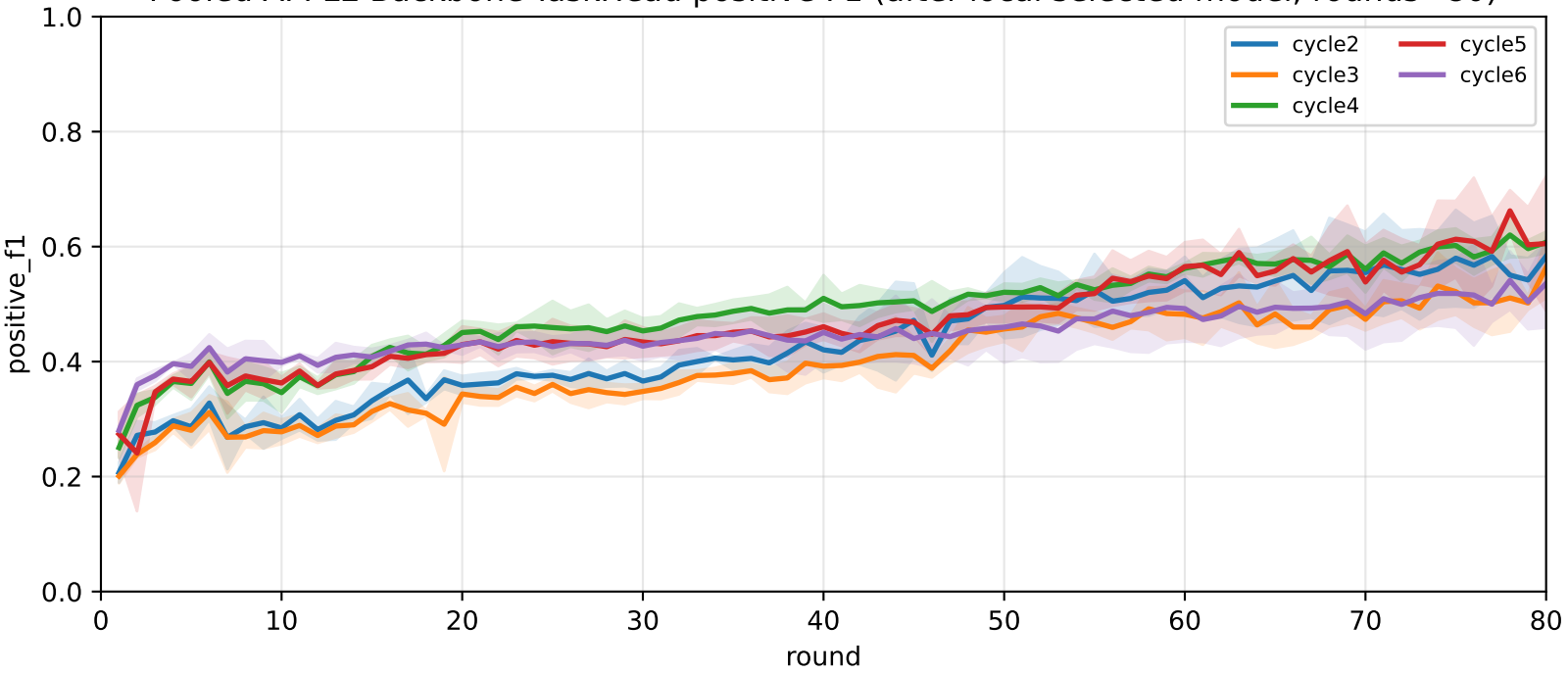
Pooled GCFL+ positive F1



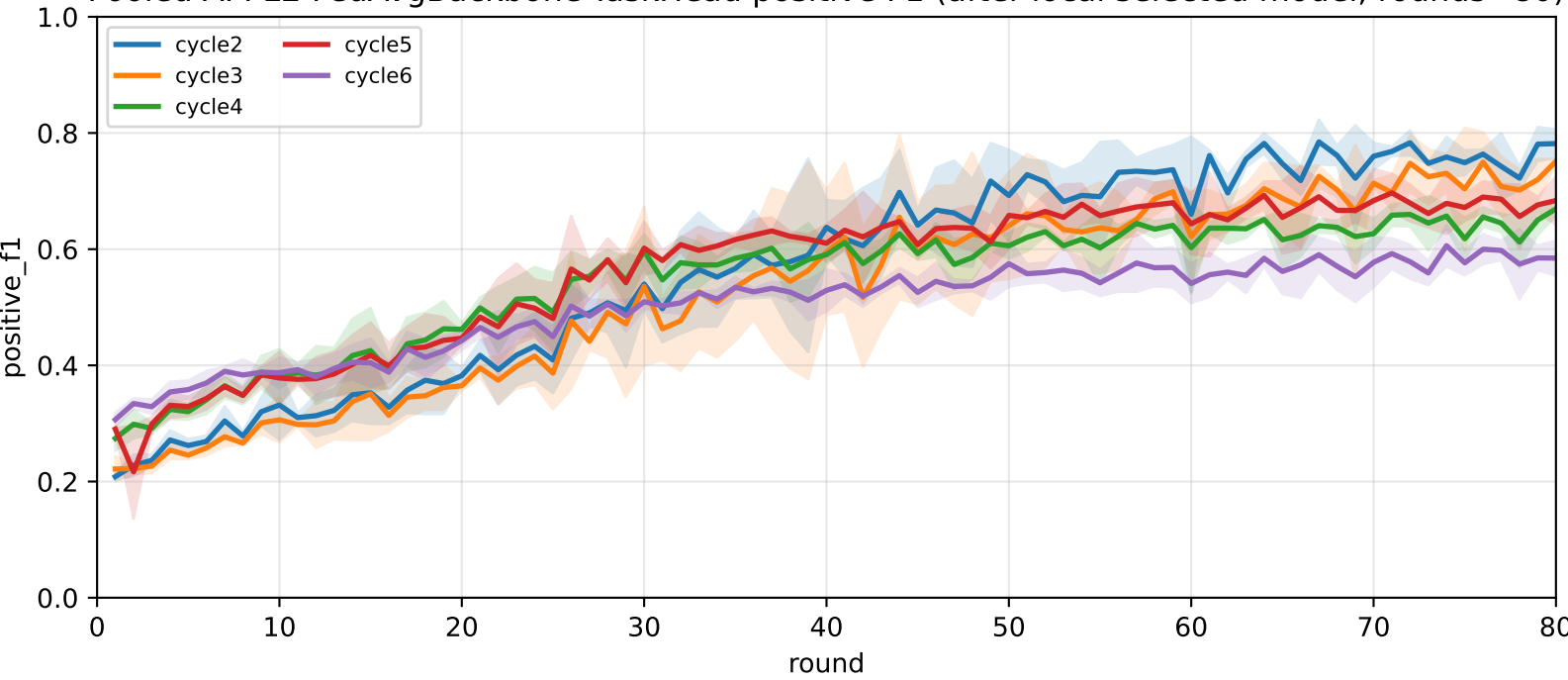
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



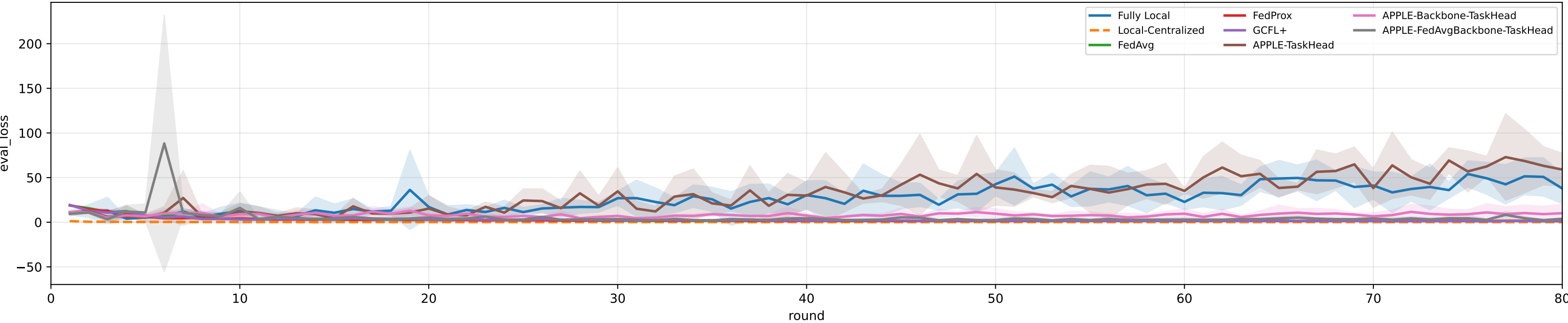
Controlled label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

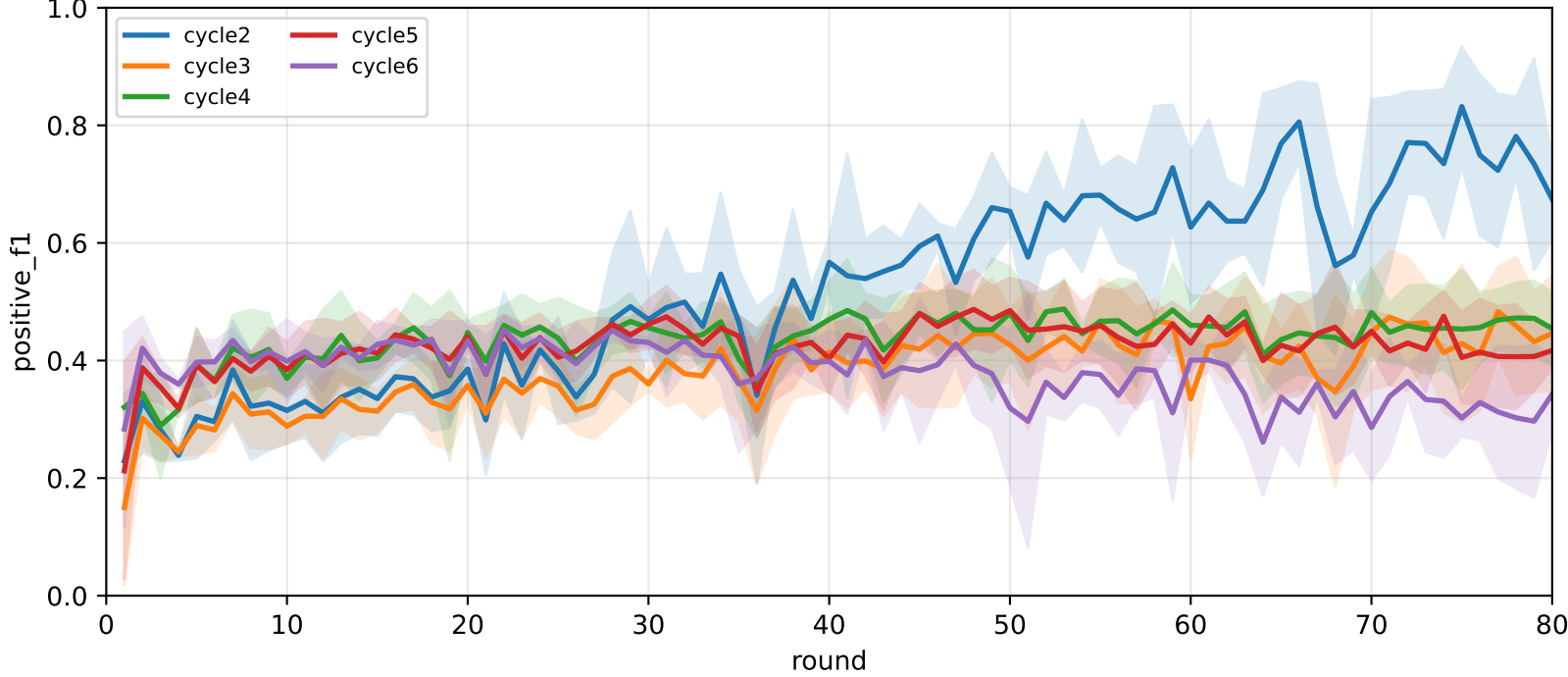
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	28.6% ± 0.7%	28.6% ± 0.7%	23.4% ± 1.7%	29.1% ± 1.2%	30.2% ± 0.8%	28.5% ± 1.4%	32.0% ± 1.3%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	<u>59.3% ± 1.6%</u>	<u>59.8% ± 1.5%</u>	59.5% ± 1.4%	<u>64.5% ± 4.3%</u>	<u>61.3% ± 3.0%</u>	60.3% ± 2.8%	<u>53.5% ± 2.6%</u>
FedProx	57.5% ± 0.6%	58.2% ± 0.3%	<u>61.0% ± 3.2%</u>	59.8% ± 3.6%	60.5% ± 1.3%	58.9% ± 1.9%	51.0% ± 1.7%
GCFL+	57.7% ± 2.4%	58.2% ± 2.7%	56.2% ± 2.0%	60.7% ± 5.9%	60.6% ± 1.1%	<u>62.5% ± 4.5%</u>	51.3% ± 0.7%
APPLE-TaskHead	33.0% ± 5.1%	33.1% ± 5.1%	27.1% ± 4.7%	34.0% ± 5.2%	34.7% ± 6.8%	33.5% ± 5.1%	36.0% ± 3.7%
APPLE-Backbone-TaskHead	36.6% ± 7.3%	36.6% ± 7.3%	31.2% ± 8.1%	37.9% ± 8.5%	36.7% ± 6.4%	36.4% ± 6.6%	40.8% ± 6.9%
APPLE-FedAvgBackbone-TaskHead	57.9% ± 3.6%	58.7% ± 3.8%	59.4% ± 7.1%	64.1% ± 6.8%	57.4% ± 0.9%	60.3% ± 5.8%	52.2% ± 2.8%

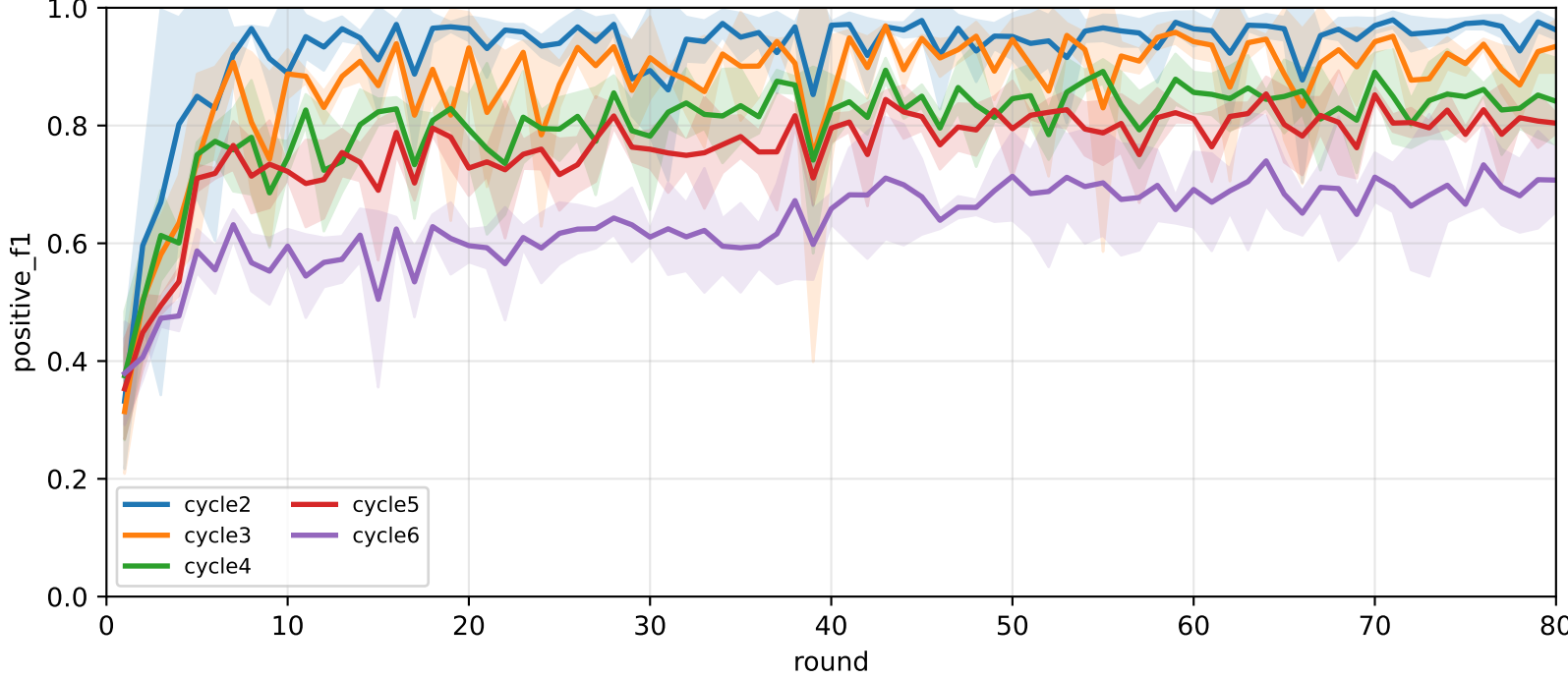
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



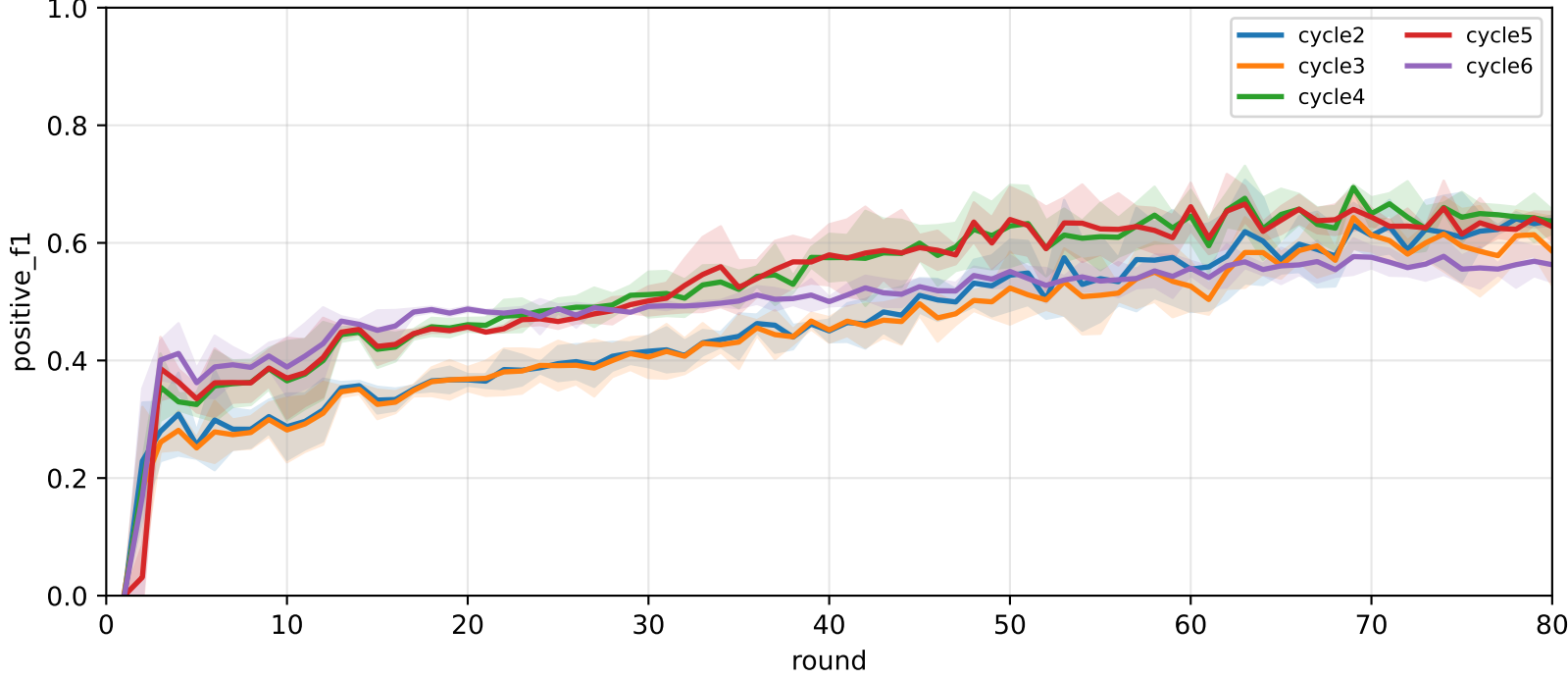
Pooled Fully Local positive F1



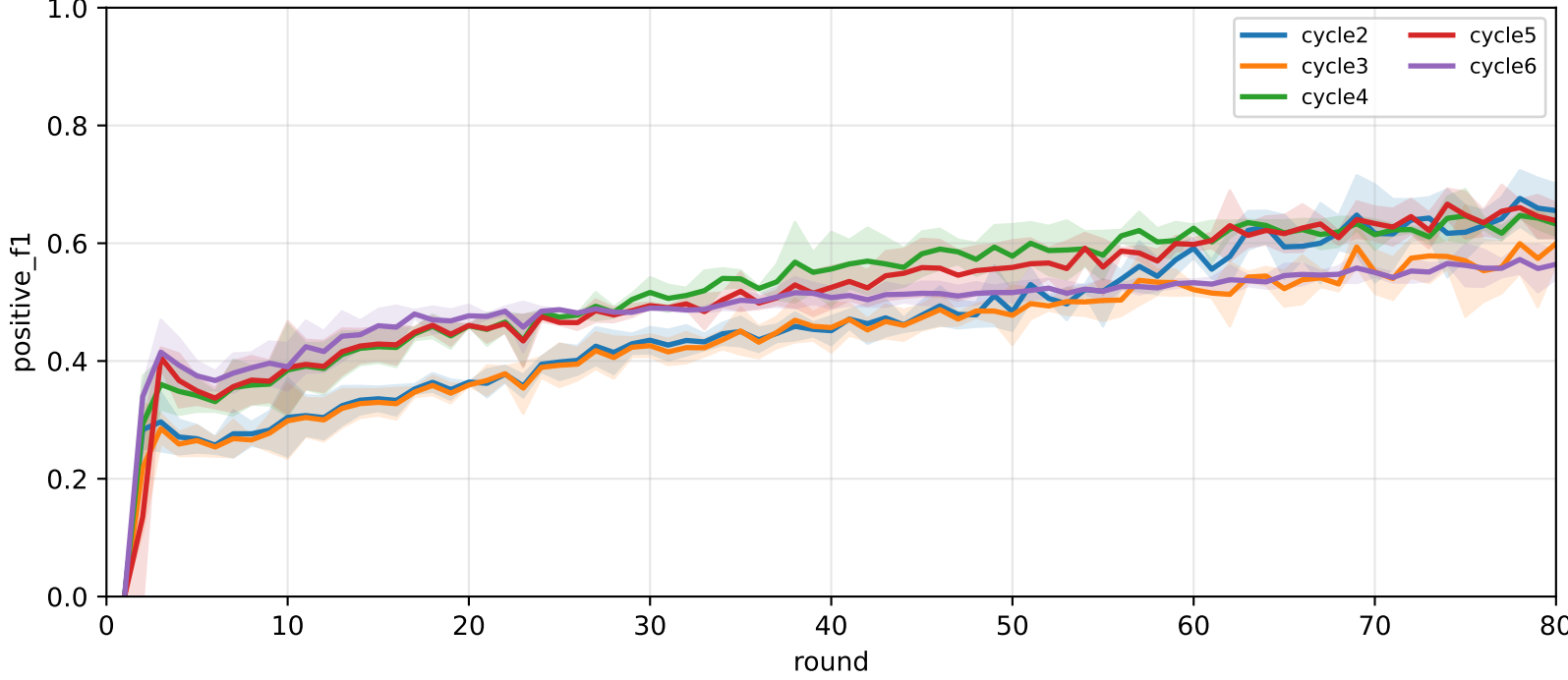
Pooled Local-Centralized positive F1



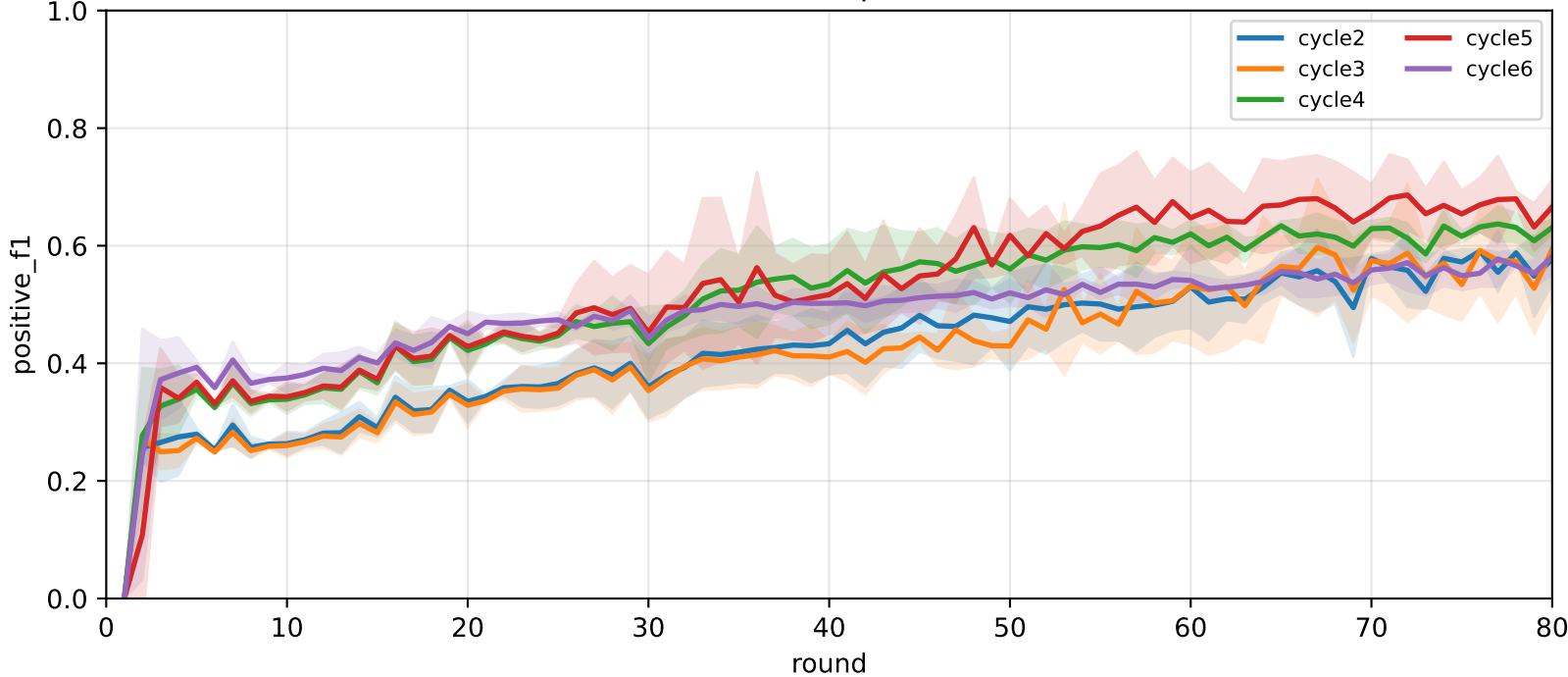
Pooled FedAvg positive F1



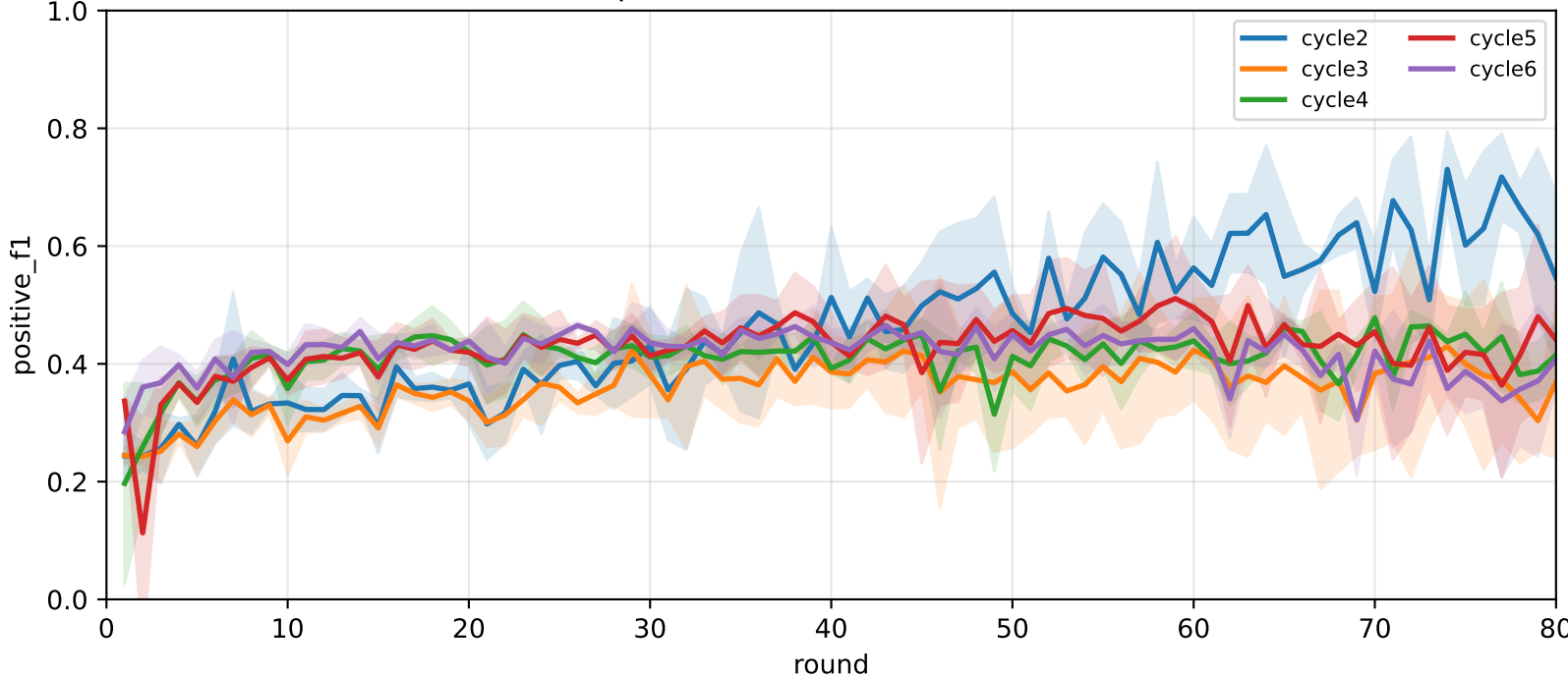
Pooled FedProx positive F1



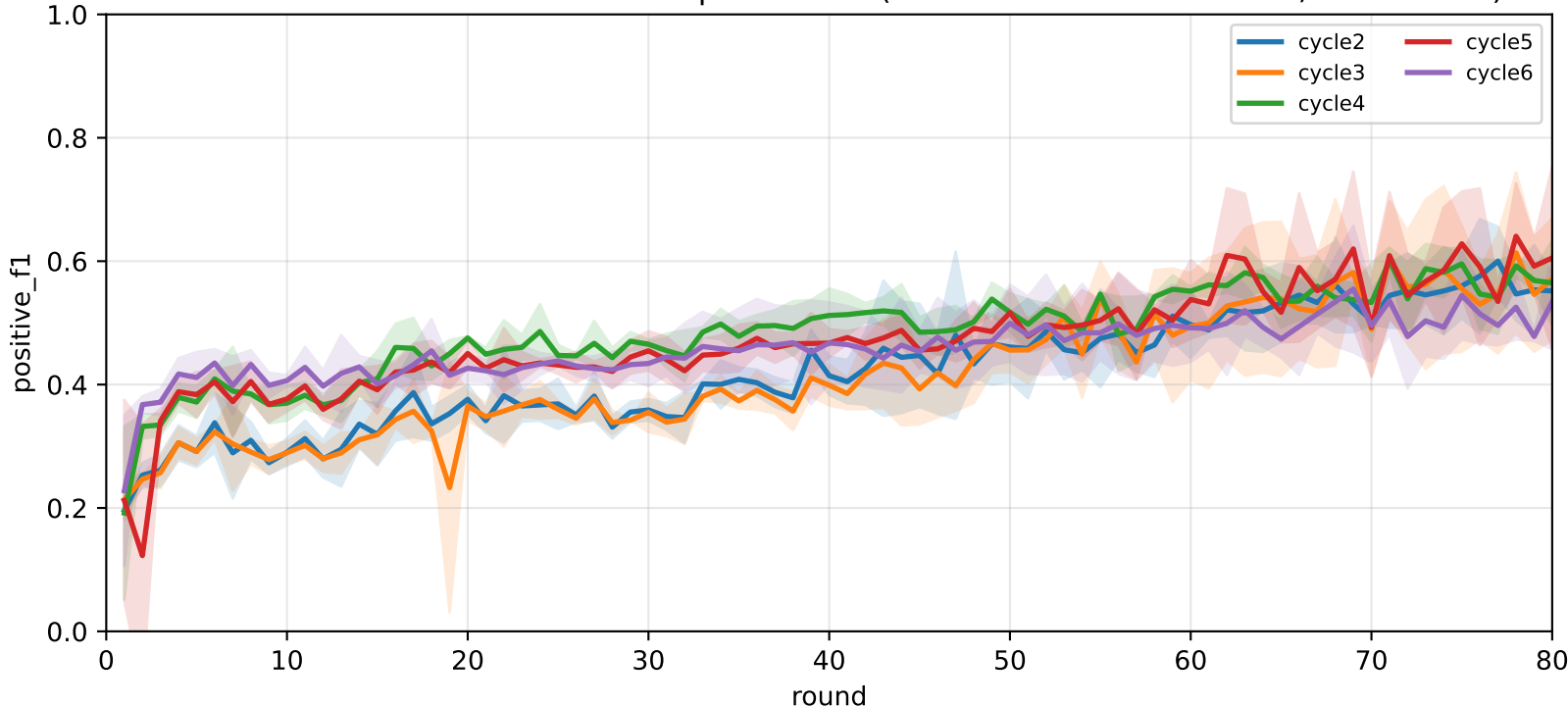
Pooled GCFL+ positive F1



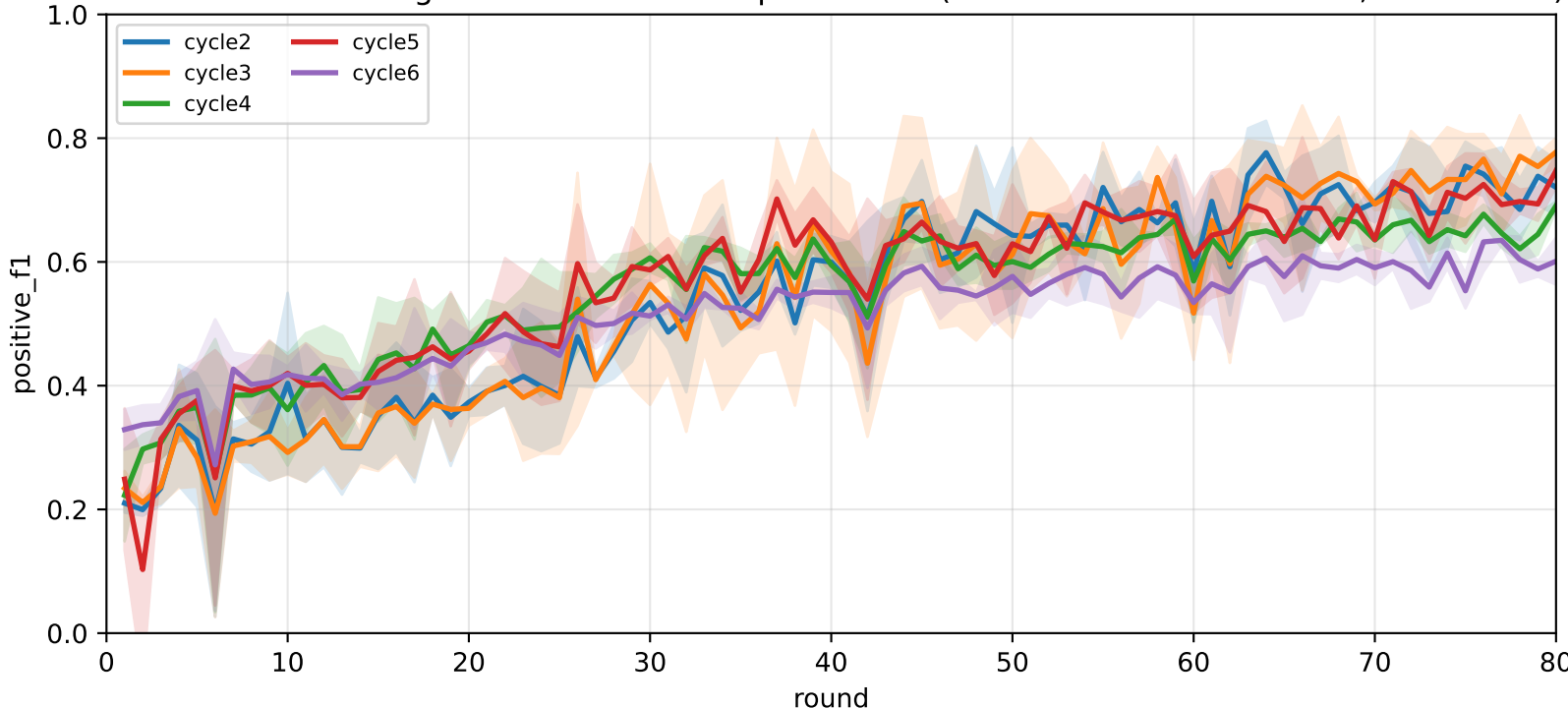
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



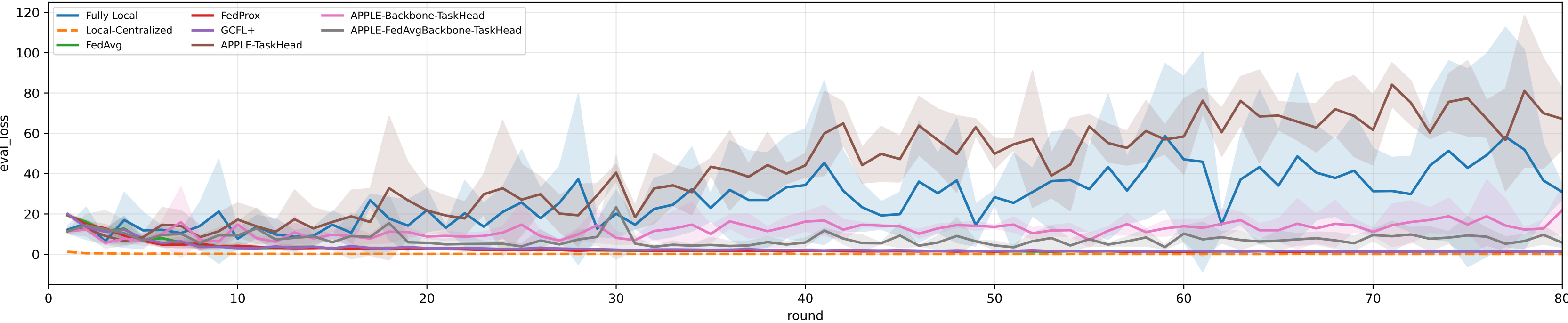
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

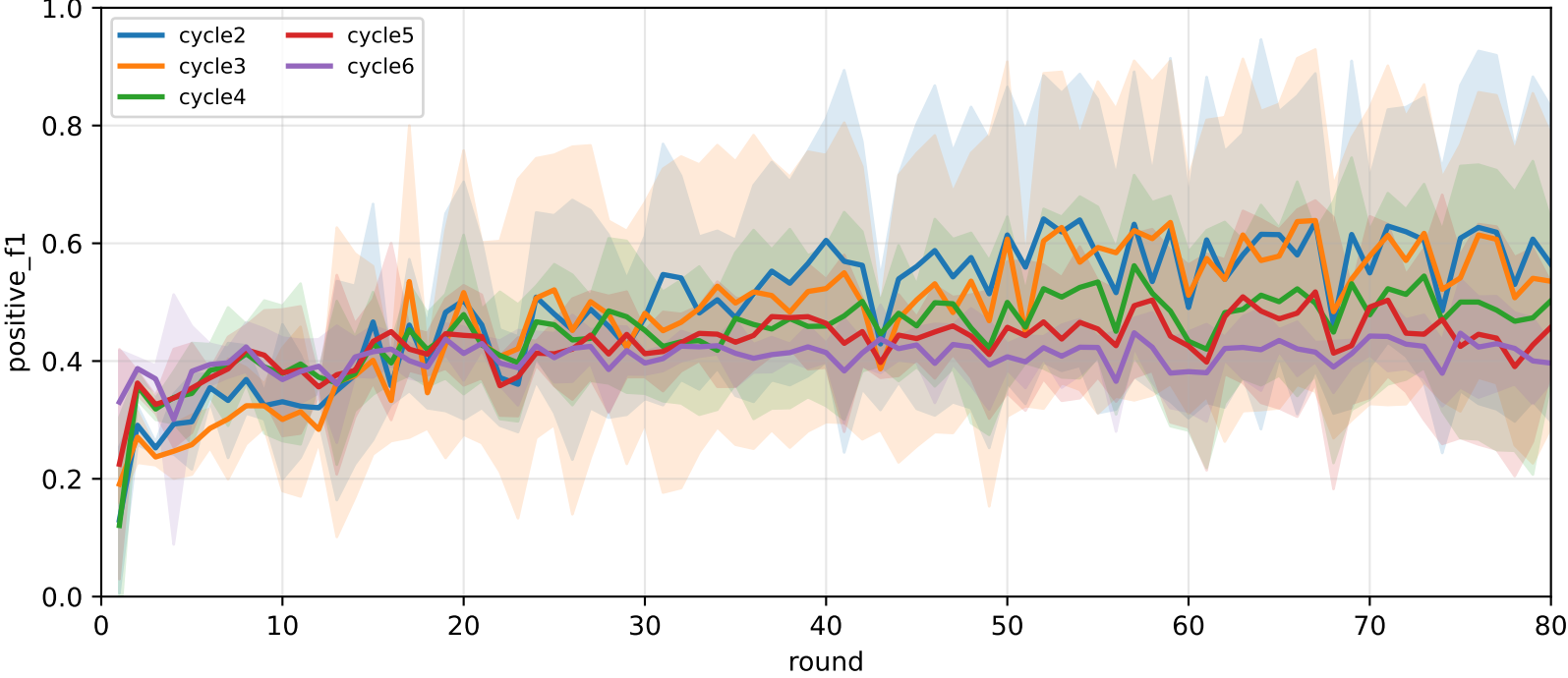
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	26.5% ± 4.7%	26.6% ± 4.7%	22.3% ± 4.2%	27.0% ± 7.0%	26.9% ± 5.3%	27.0% ± 4.1%	29.5% ± 3.2%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	59.6% ± 1.7%	60.2% ± 1.6%	60.7% ± 1.9%	<u>64.7% ± 5.0%</u>	<u>61.9% ± 2.6%</u>	60.2% ± 1.9%	53.3% ± 3.1%
FedProx	58.0% ± 1.0%	58.8% ± 0.6%	62.4% ± 2.8%	60.8% ± 4.0%	60.7% ± 1.1%	59.2% ± 1.1%	50.9% ± 2.3%
GCFL+	57.3% ± 2.4%	57.8% ± 2.7%	55.9% ± 2.0%	60.9% ± 5.7%	59.8% ± 1.7%	61.3% ± 4.1%	51.1% ± 0.8%
APPLE-TaskHead	33.6% ± 0.3%	33.6% ± 0.4%	29.3% ± 1.4%	32.6% ± 1.5%	33.9% ± 0.5%	35.1% ± 1.4%	37.0% ± 0.9%
APPLE-Backbone-TaskHead	36.0% ± 7.0%	36.0% ± 6.9%	30.4% ± 7.7%	36.1% ± 7.7%	36.8% ± 8.1%	36.4% ± 6.6%	40.1% ± 4.8%
APPLE-FedAvgBackbone-TaskHead	<u>60.9% ± 8.2%</u>	<u>61.9% ± 8.5%</u>	<u>67.5% ± 12.8%</u>	60.5% ± 12.4%	60.1% ± 6.9%	<u>67.7% ± 5.1%</u>	<u>53.8% ± 6.2%</u>

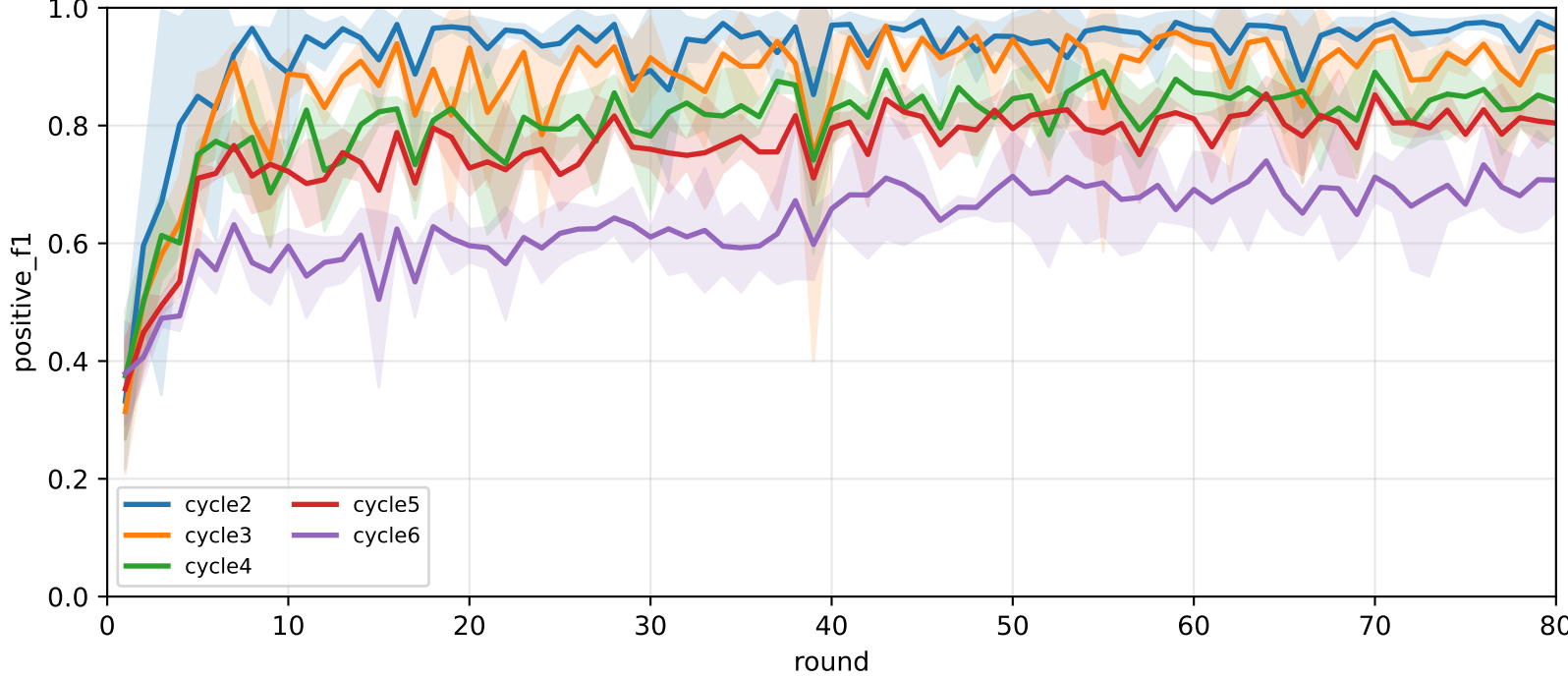
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



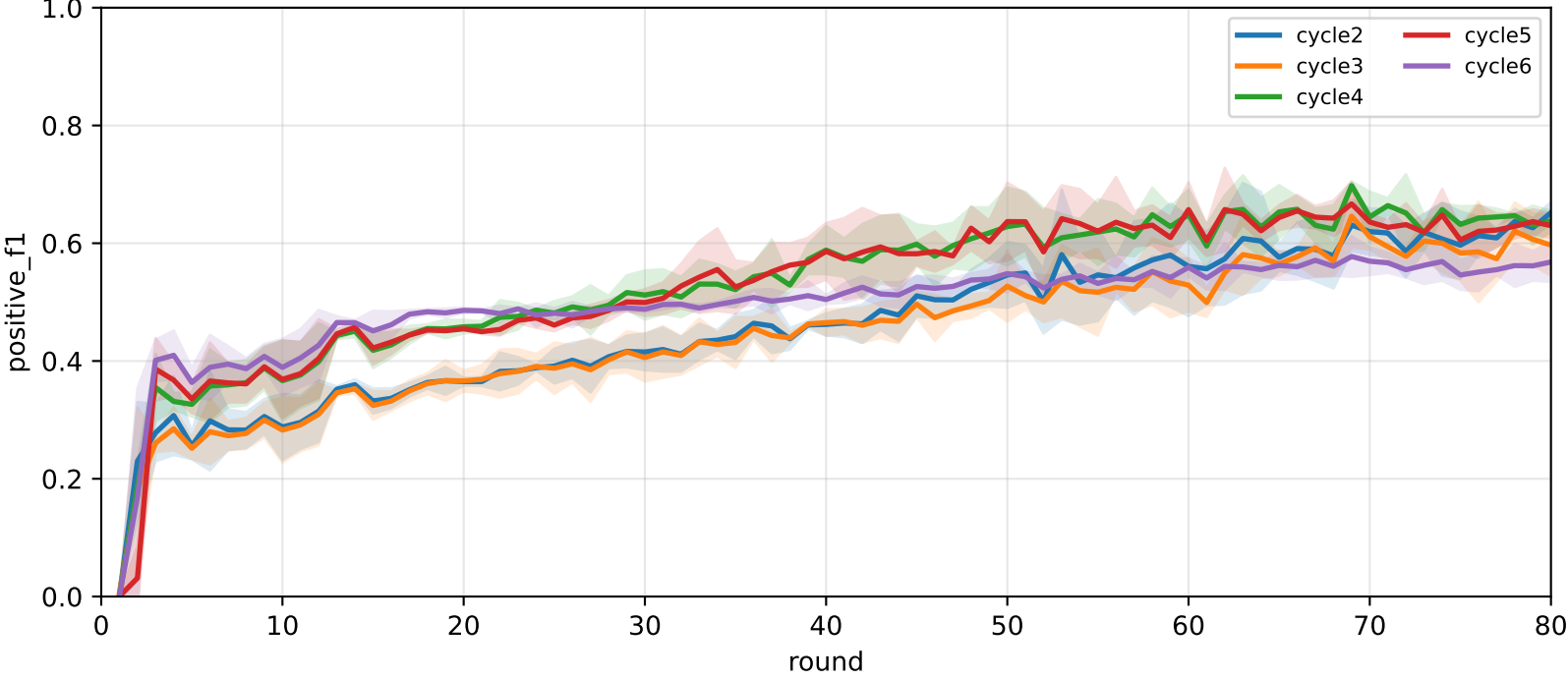
Pooled Fully Local positive F1



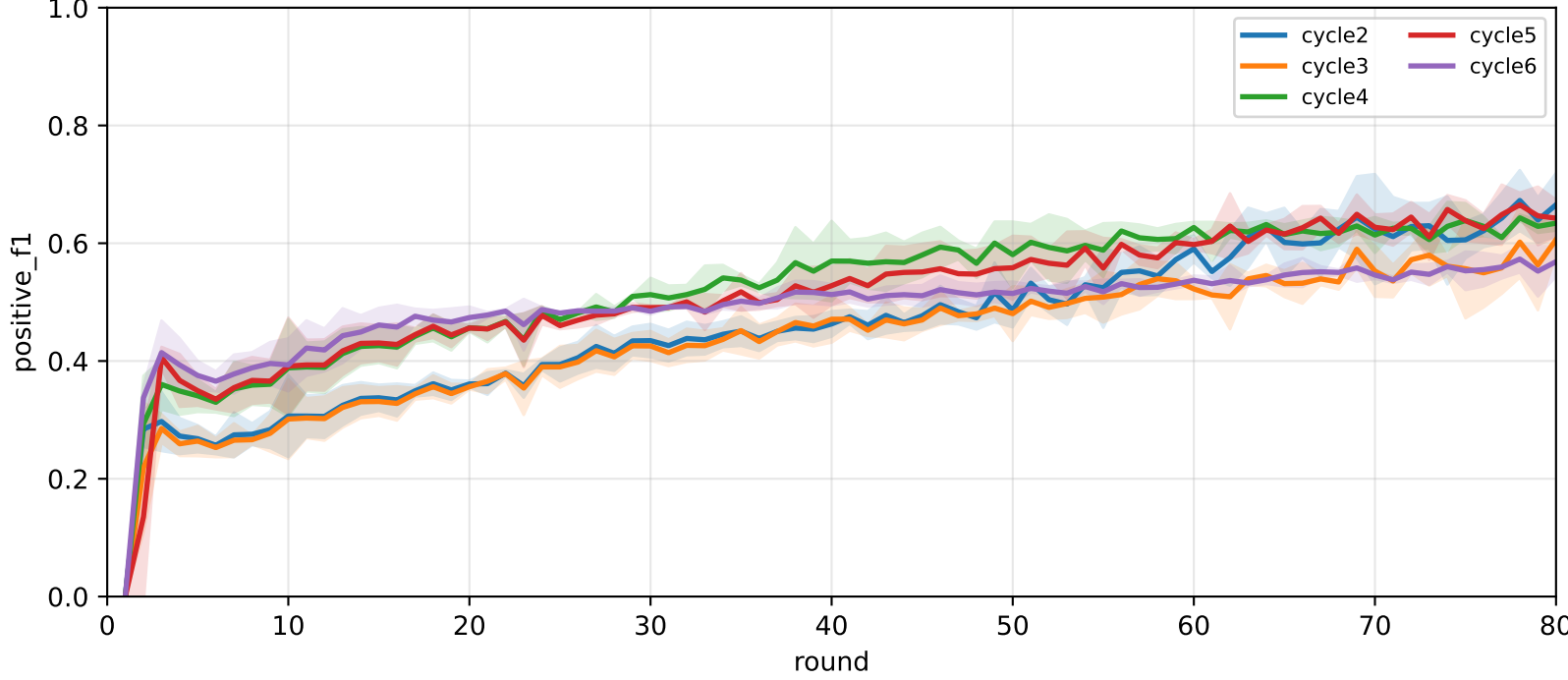
Pooled Local-Centralized positive F1



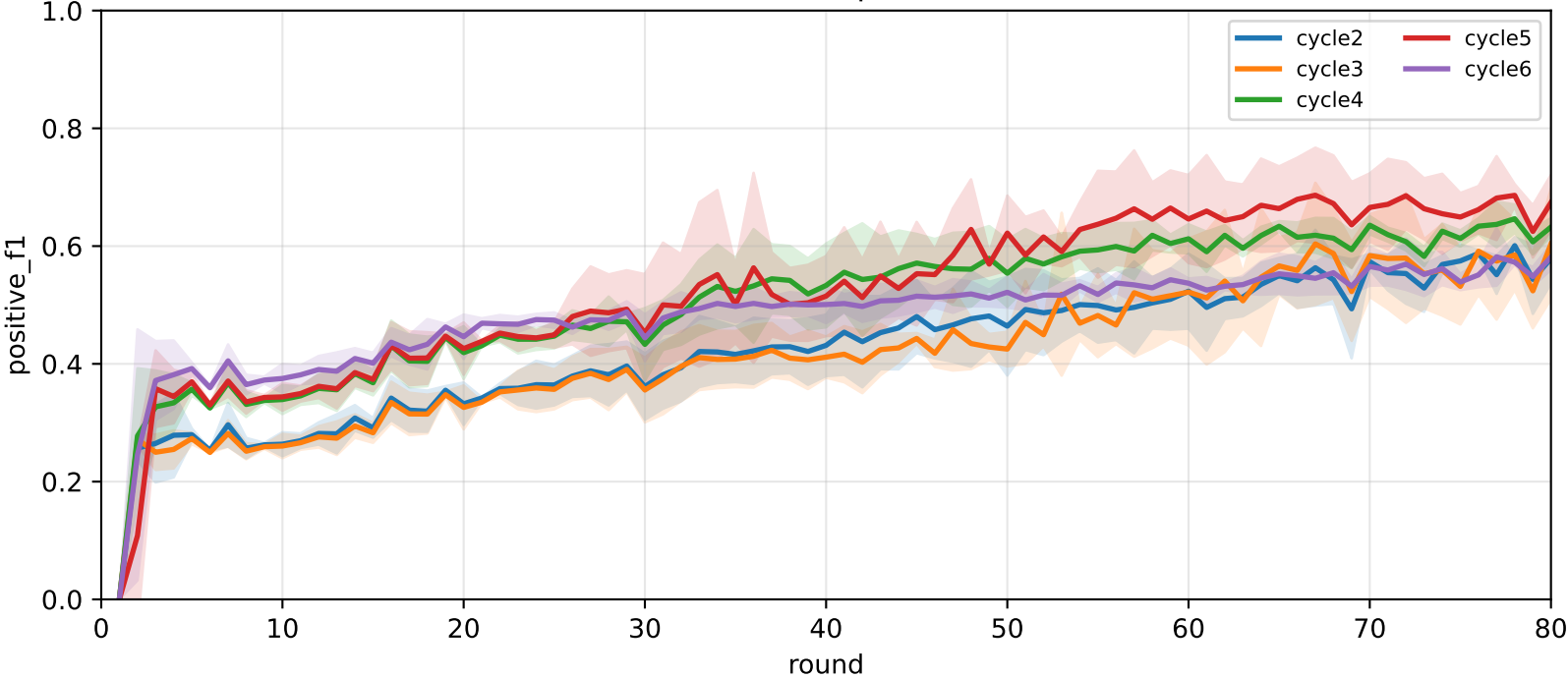
Pooled FedAvg positive F1



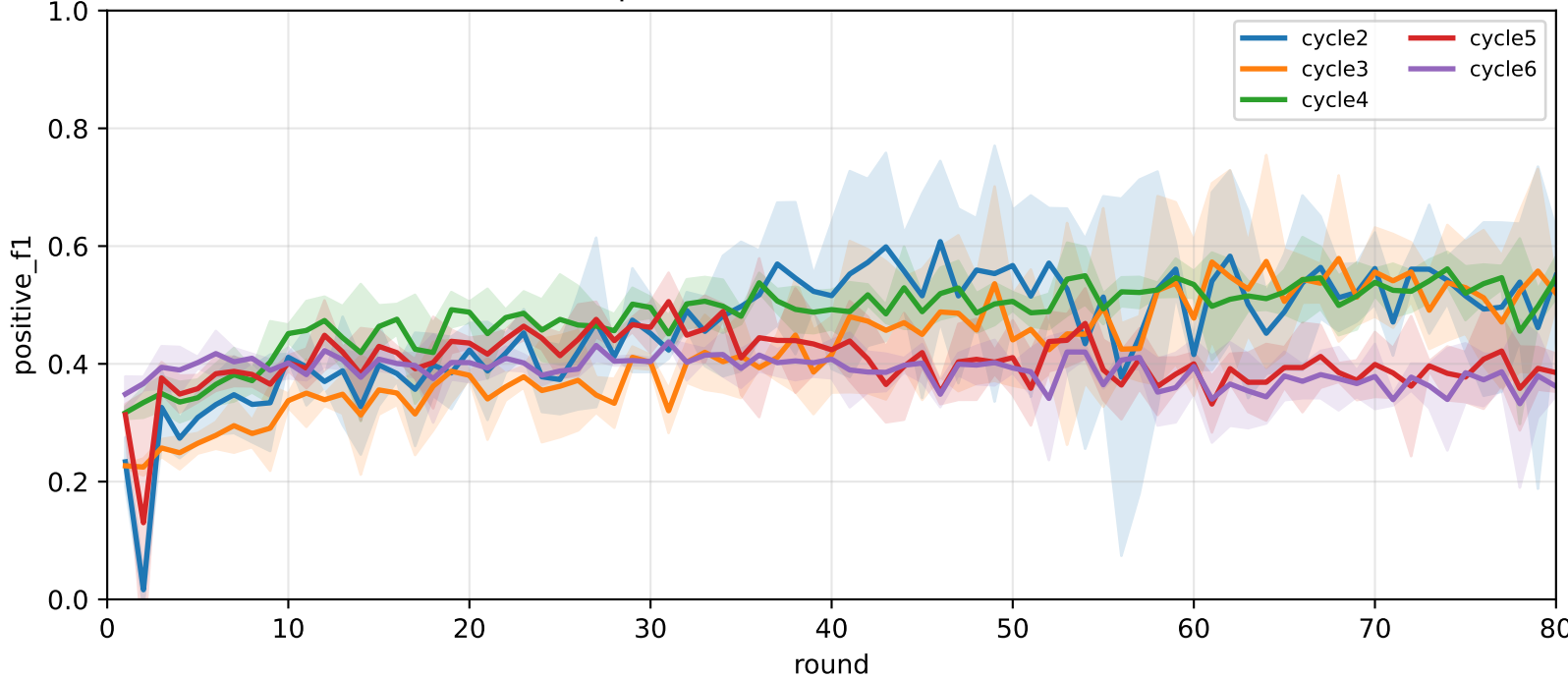
Pooled FedProx positive F1



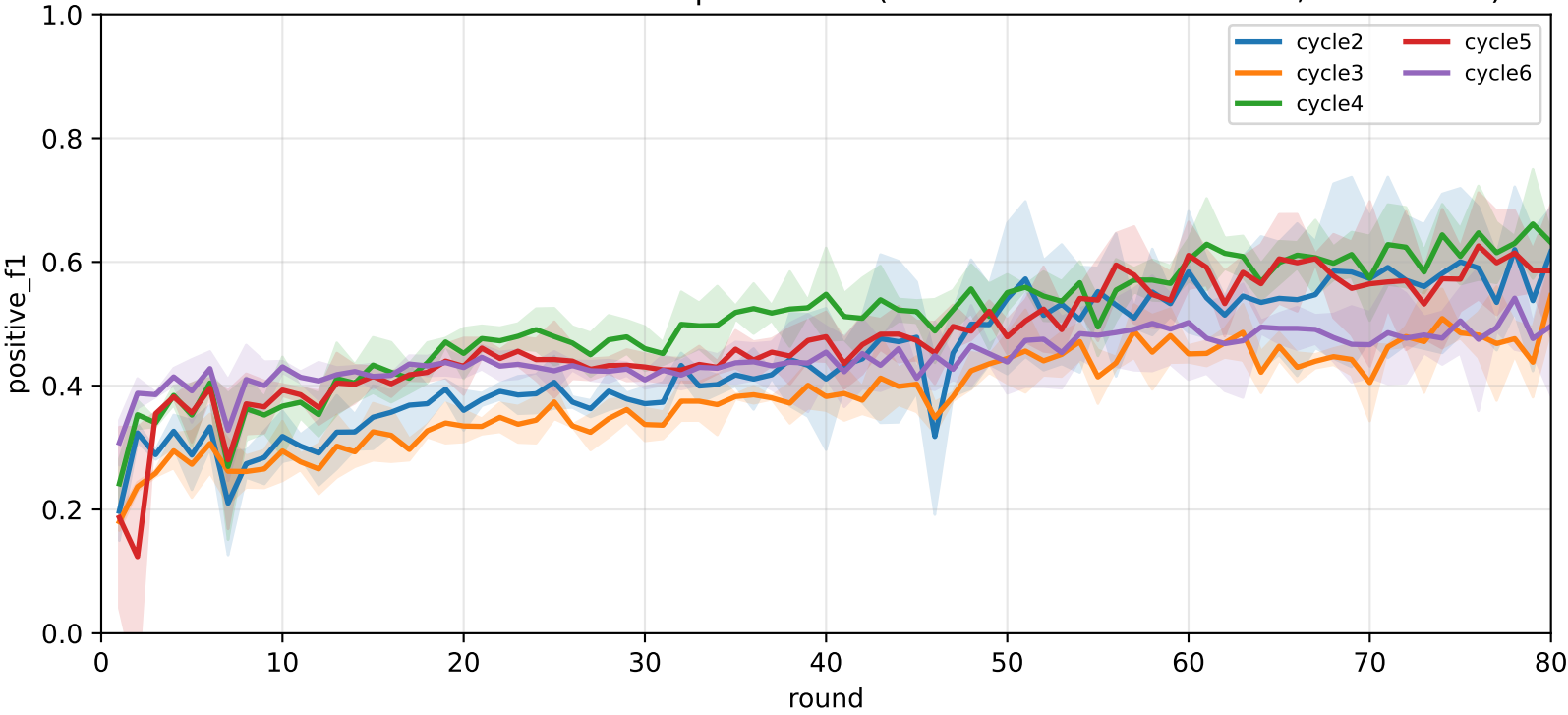
Pooled GCFL+ positive F1



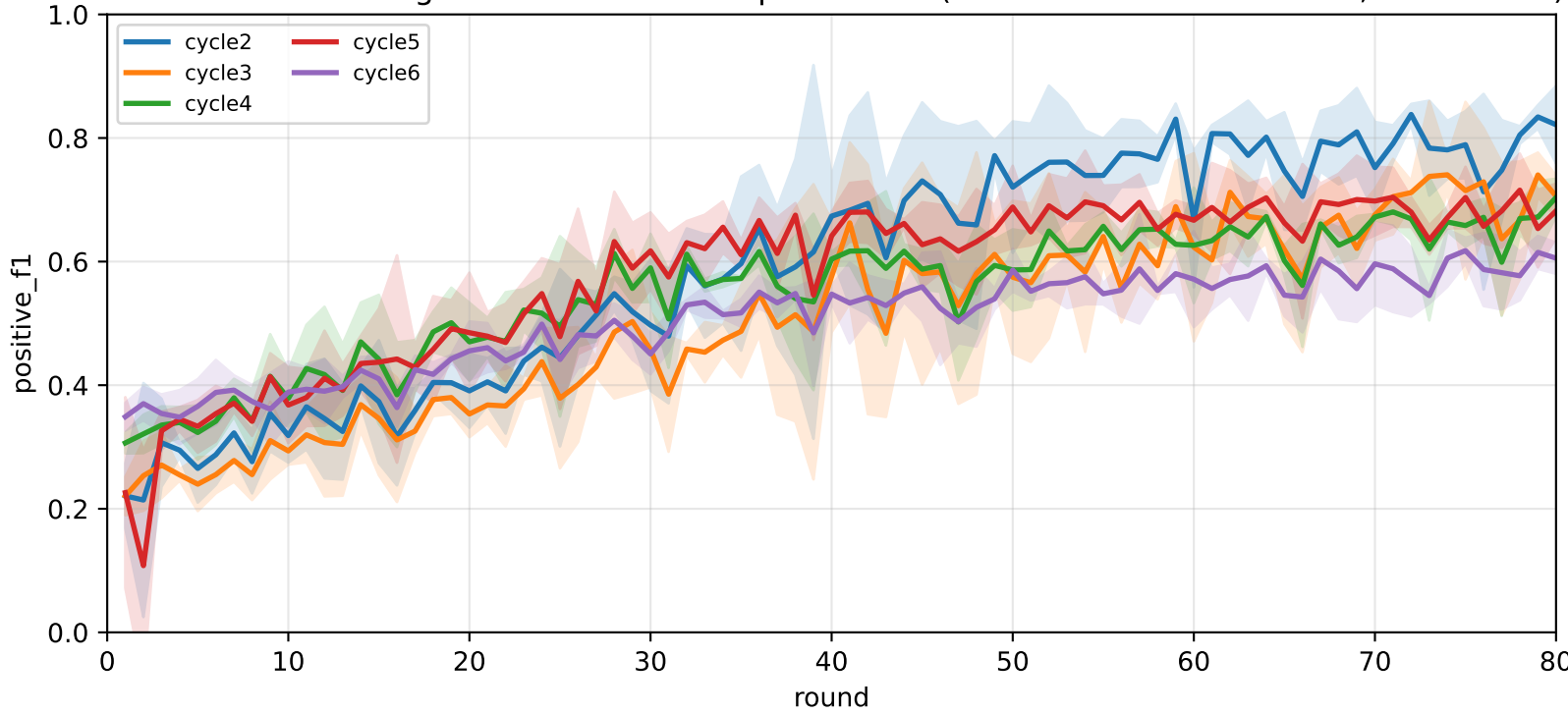
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



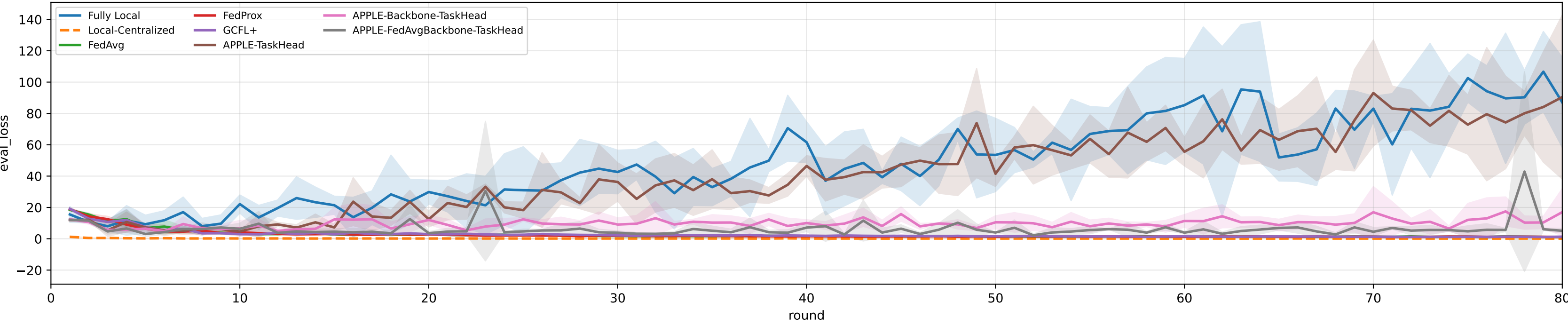
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

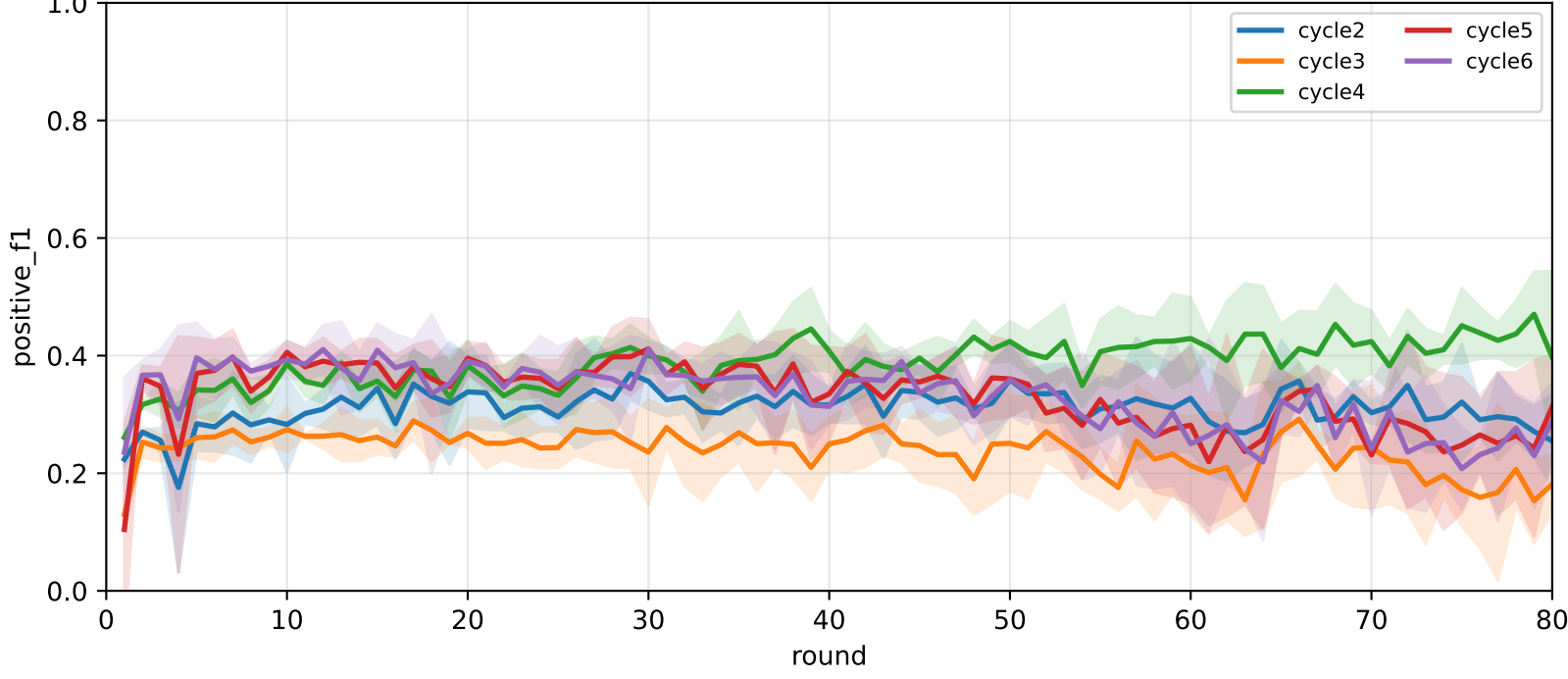
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.5% ± 1.5%	27.5% ± 1.4%	23.1% ± 1.8%	28.2% ± 0.7%	26.4% ± 1.8%	28.3% ± 1.7%	31.5% ± 2.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	<u>58.9% ± 2.1%</u>	<u>59.5% ± 2.1%</u>	60.3% ± 2.1%	<u>64.8% ± 5.2%</u>	<u>60.6% ± 1.9%</u>	59.1% ± 3.0%	<u>53.0% ± 2.9%</u>
FedProx	57.1% ± 1.2%	57.9% ± 0.9%	<u>61.6% ± 3.0%</u>	59.9% ± 4.1%	59.6% ± 2.1%	58.0% ± 2.3%	50.5% ± 2.1%
GCFL+	57.3% ± 2.7%	57.8% ± 3.0%	55.8% ± 2.0%	60.5% ± 5.9%	59.9% ± 1.9%	<u>61.8% ± 4.8%</u>	51.2% ± 0.8%
APPLE-TaskHead	30.4% ± 3.2%	30.4% ± 3.2%	26.1% ± 2.7%	30.8% ± 3.4%	30.0% ± 3.9%	30.9% ± 3.5%	33.9% ± 3.3%
APPLE-Backbone-TaskHead	35.9% ± 6.0%	35.8% ± 6.0%	31.1% ± 6.2%	36.8% ± 7.2%	35.7% ± 5.7%	36.4% ± 5.8%	39.1% ± 5.3%
APPLE-FedAvgBackbone-TaskHead	53.6% ± 8.9%	54.4% ± 9.6%	56.0% ± 11.7%	56.5% ± 15.5%	51.5% ± 8.9%	58.3% ± 10.5%	49.9% ± 3.6%

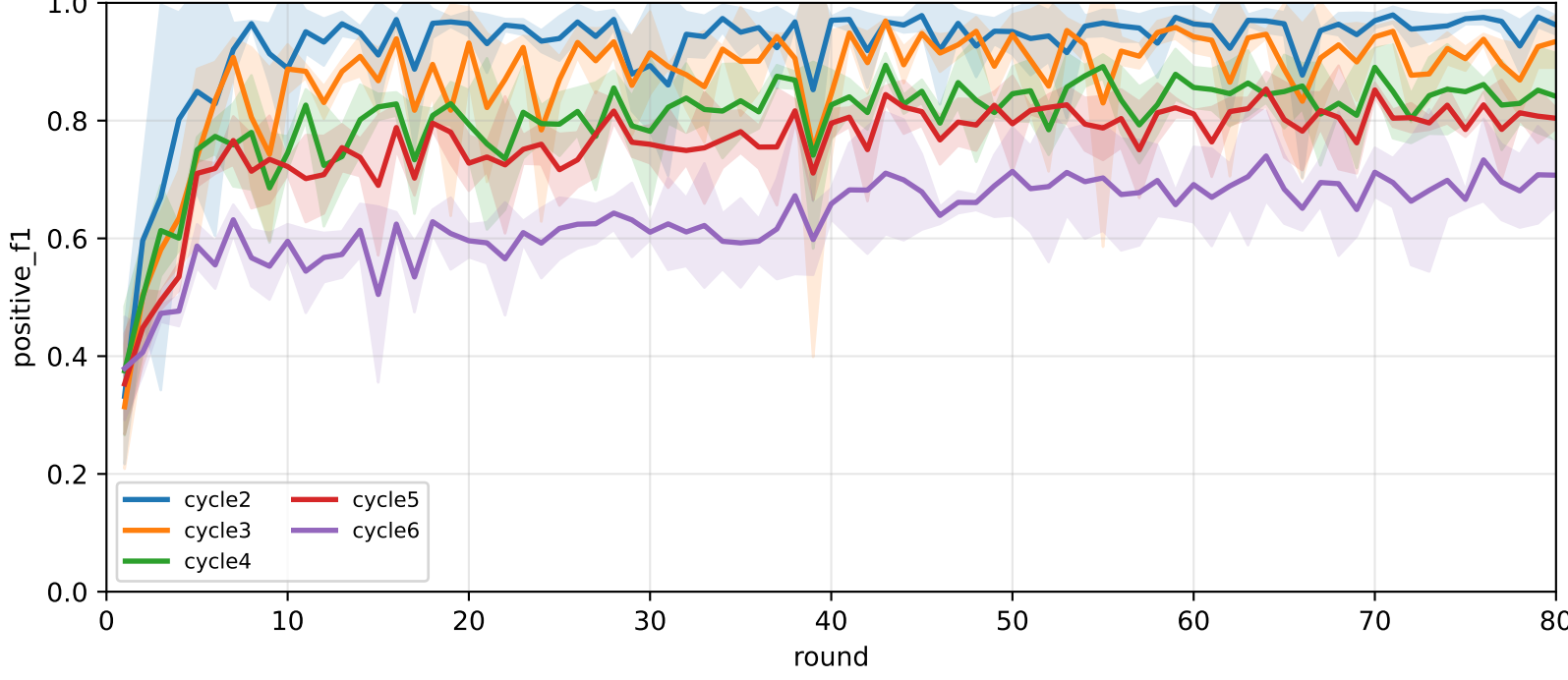
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



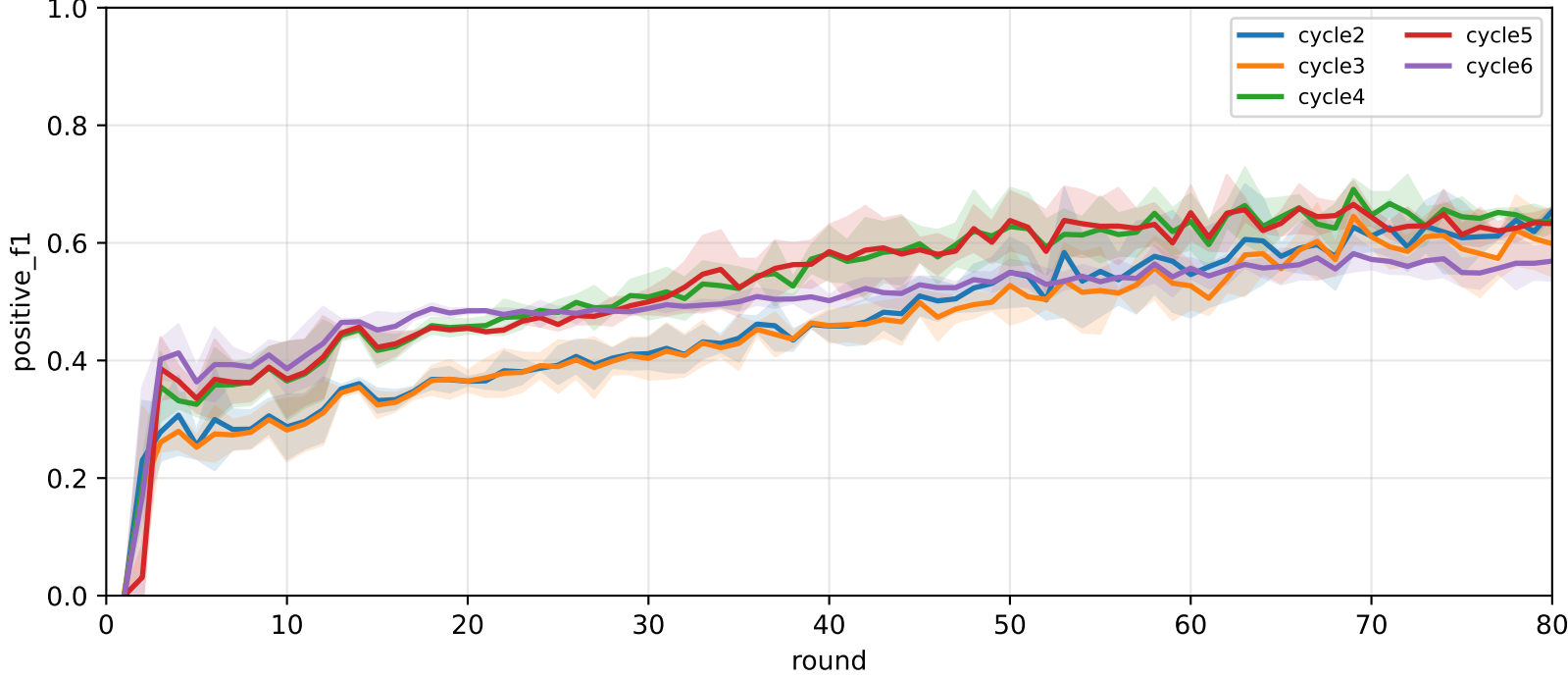
Pooled Fully Local positive F1



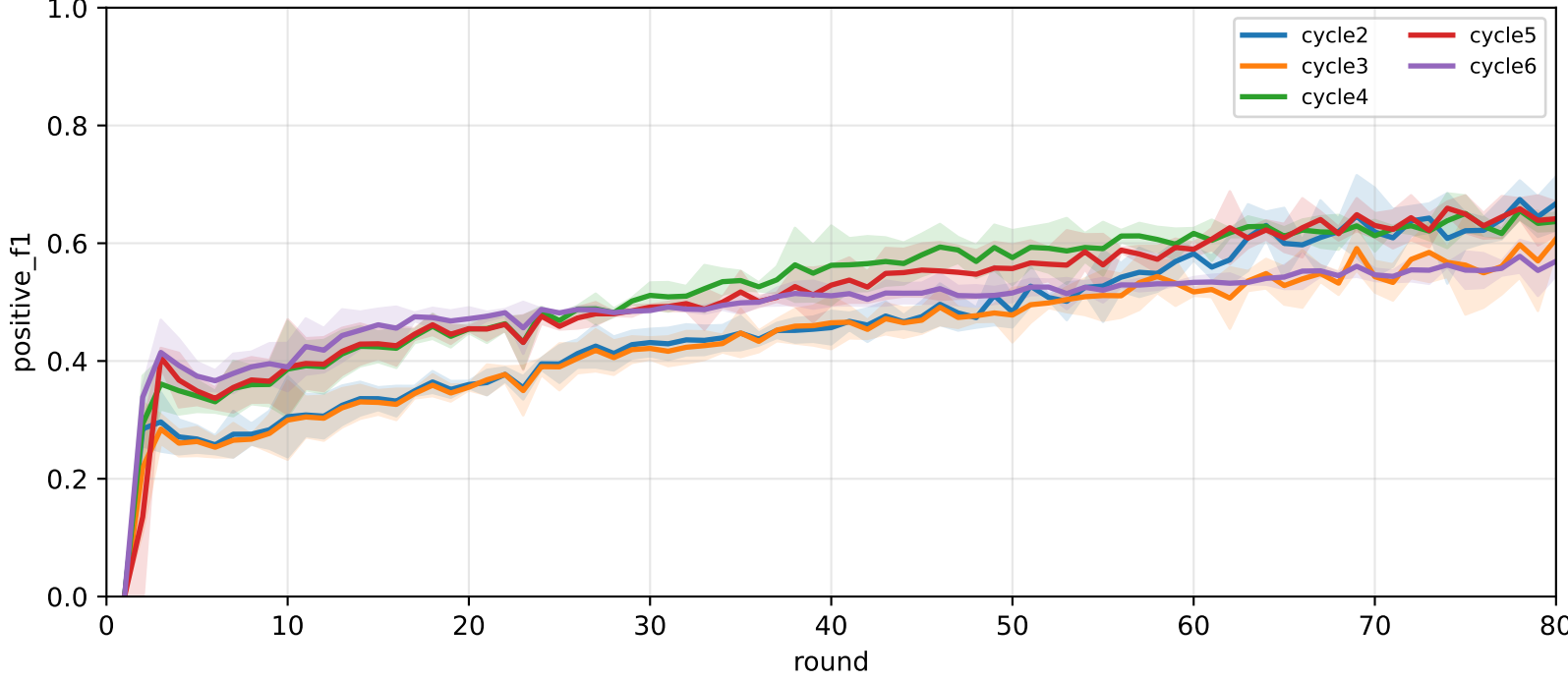
Pooled Local-Centralized positive F1



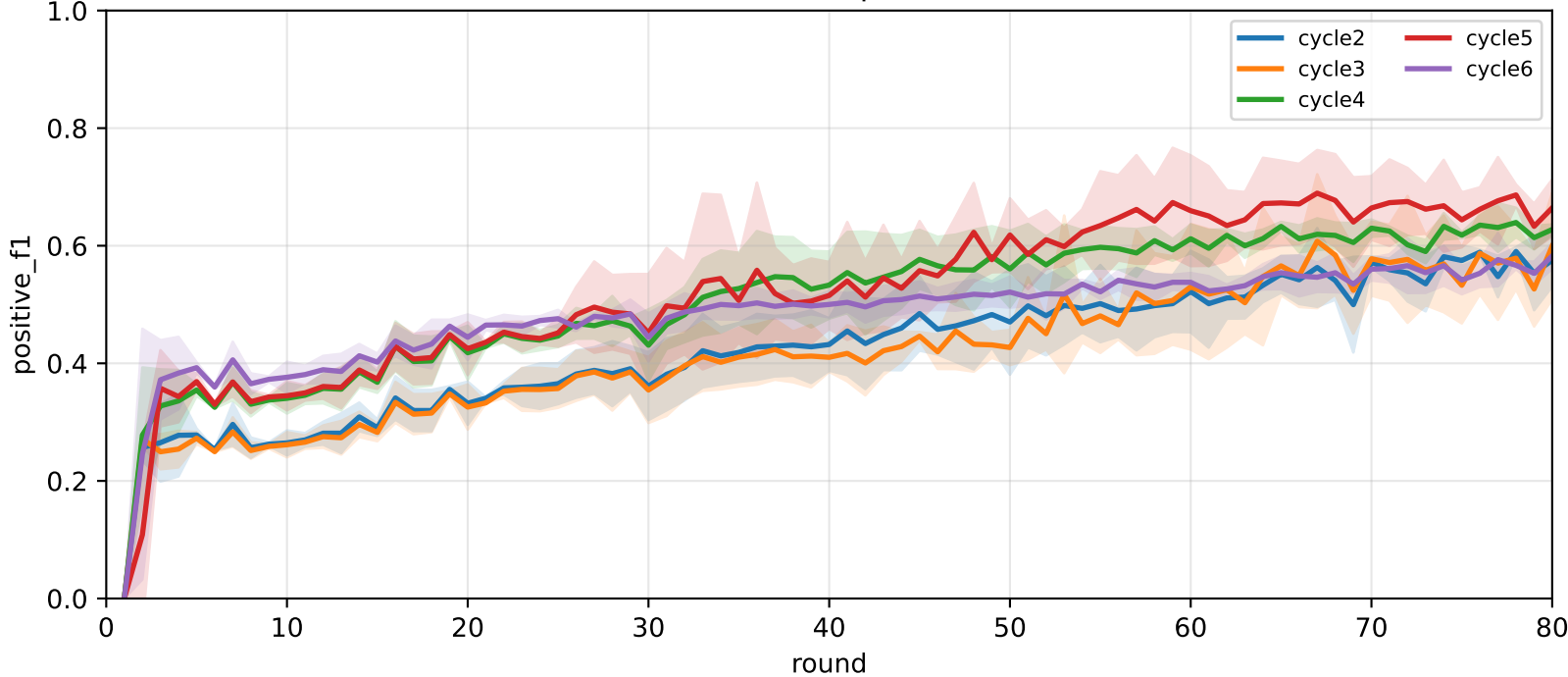
Pooled FedAvg positive F1



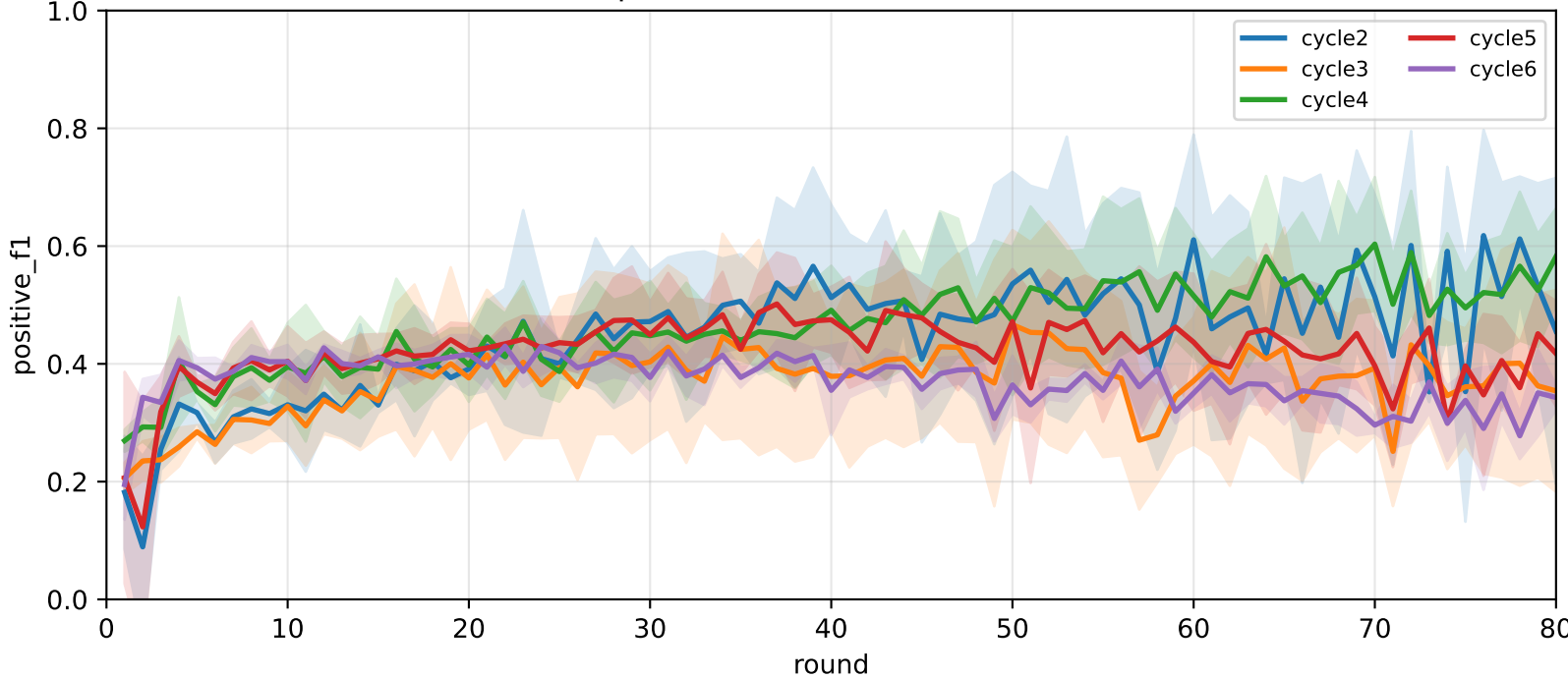
Pooled FedProx positive F1



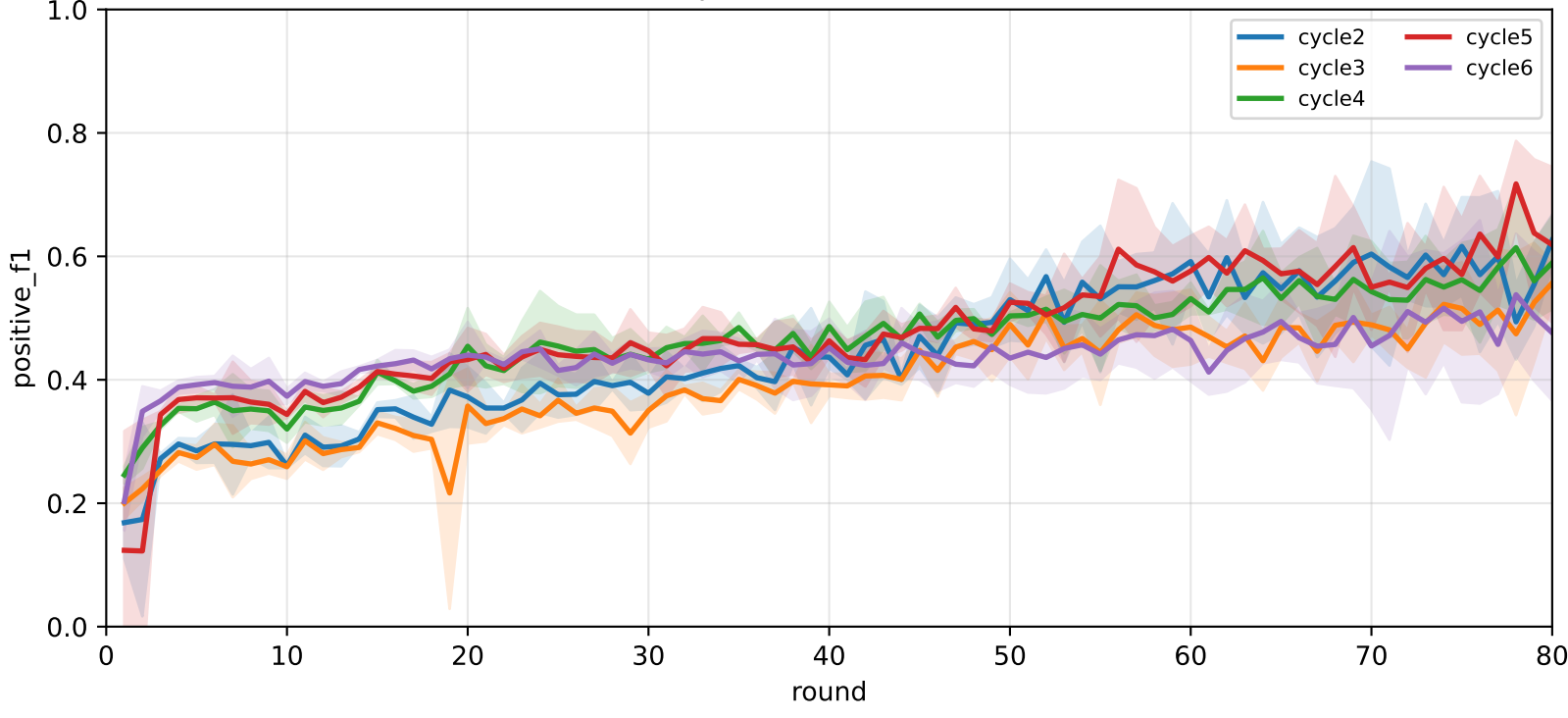
Pooled GCFL+ positive F1



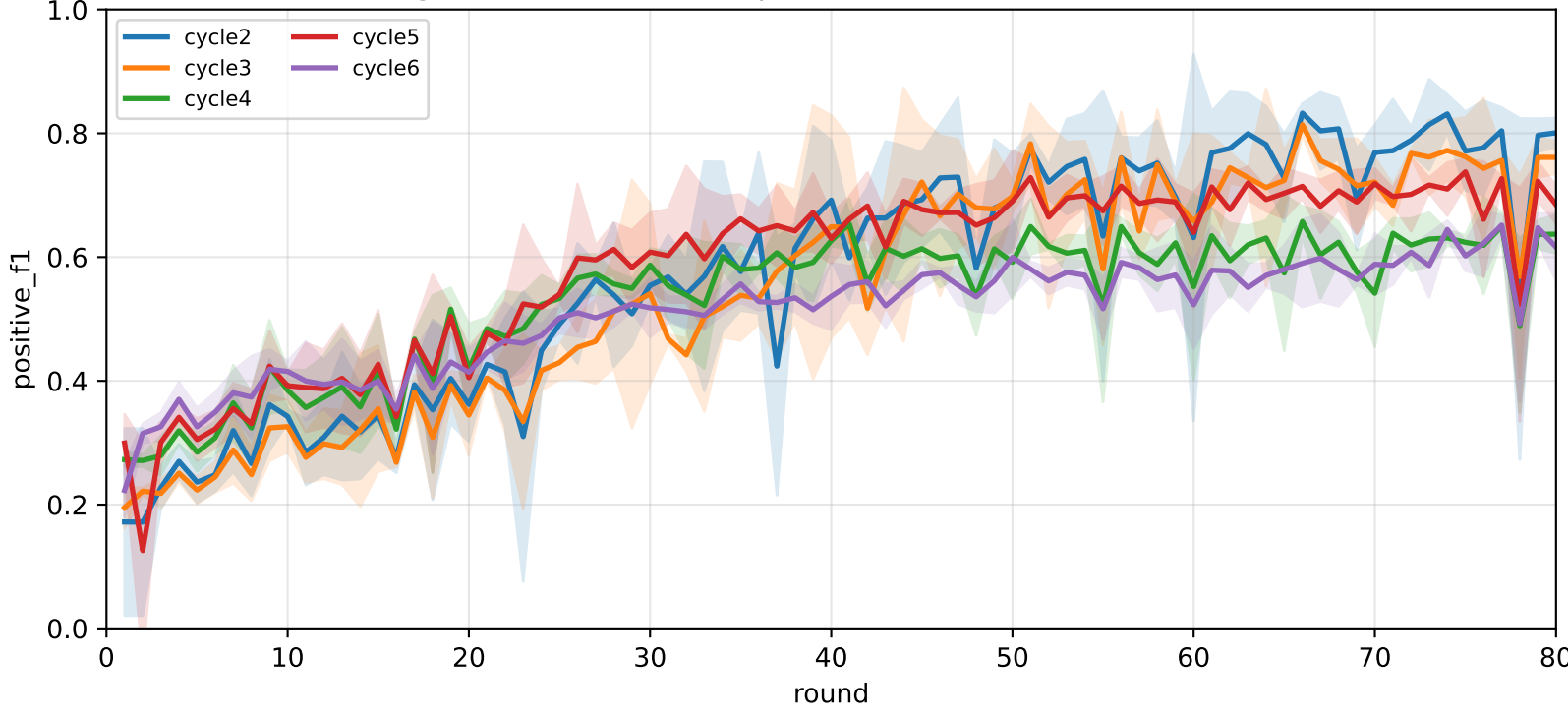
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



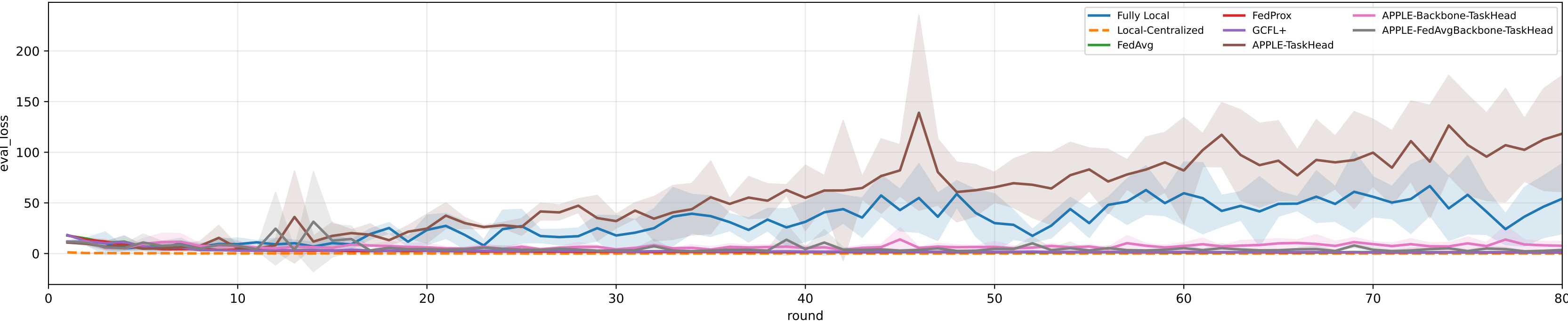
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

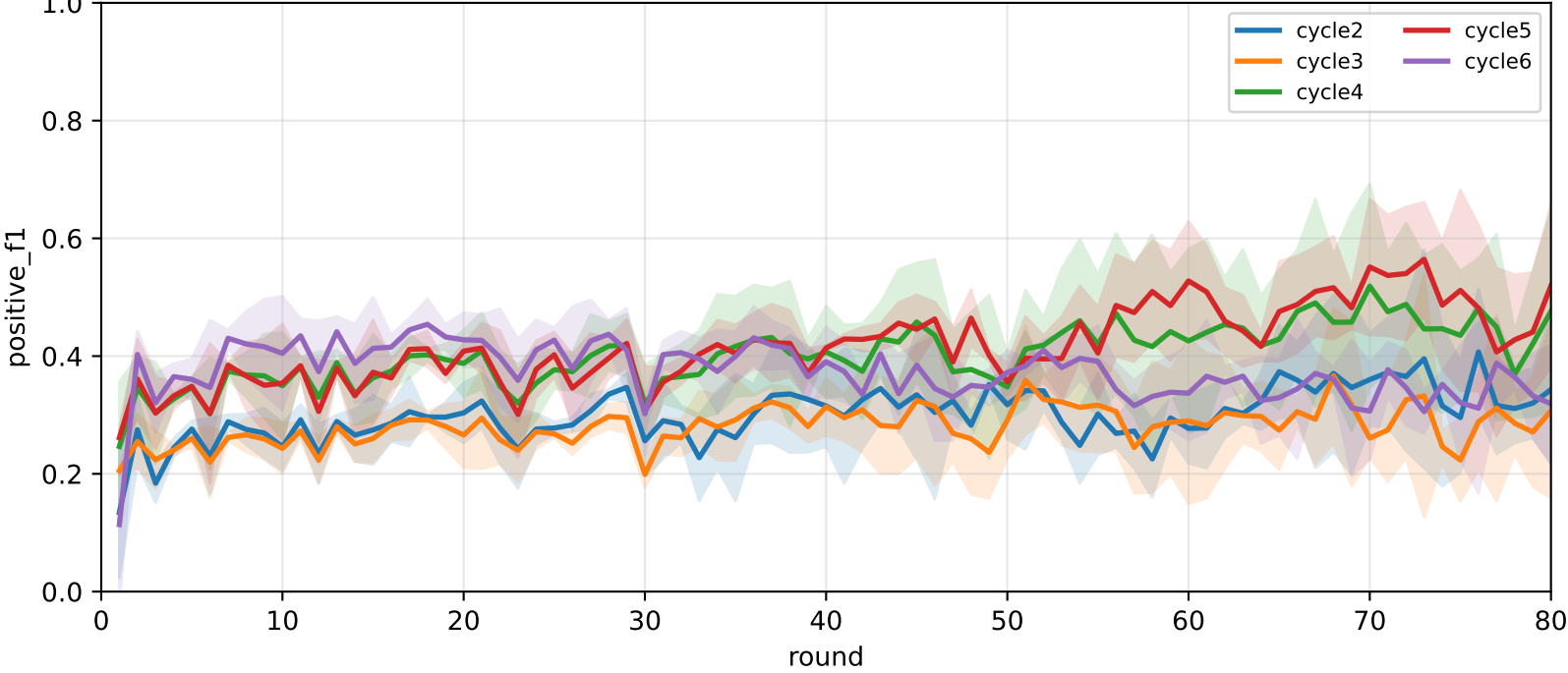
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	24.2% ± 1.1%	24.2% ± 1.1%	20.1% ± 0.7%	25.3% ± 1.1%	24.5% ± 1.6%	22.9% ± 1.3%	28.3% ± 1.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	<u>59.3% ± 1.8%</u>	59.9% ± 1.8%	60.0% ± 2.5%	<u>65.6% ± 5.4%</u>	61.2% ± 2.5%	59.6% ± 2.5%	<u>53.2% ± 3.2%</u>
FedProx	57.3% ± 1.2%	58.1% ± 0.8%	61.4% ± 2.4%	60.3% ± 4.4%	60.0% ± 1.1%	57.9% ± 1.8%	50.7% ± 2.6%
GCFL+	57.1% ± 2.5%	57.6% ± 2.7%	55.3% ± 1.8%	61.0% ± 6.0%	59.8% ± 1.4%	<u>61.1% ± 4.5%</u>	50.9% ± 0.7%
APPLE-TaskHead	26.6% ± 1.9%	26.5% ± 2.0%	22.2% ± 1.2%	27.3% ± 2.5%	27.2% ± 2.4%	25.2% ± 1.6%	30.8% ± 2.3%
APPLE-Backbone-TaskHead	35.1% ± 5.2%	35.0% ± 5.3%	29.9% ± 6.1%	35.7% ± 6.1%	35.7% ± 5.0%	34.3% ± 3.8%	39.4% ± 5.3%
APPLE-FedAvgBackbone-TaskHead	<u>59.3% ± 8.8%</u>	<u>60.4% ± 9.5%</u>	<u>64.9% ± 15.5%</u>	<u>65.6% ± 12.0%</u>	<u>61.4% ± 7.2%</u>	57.6% ± 10.2%	52.7% ± 3.8%

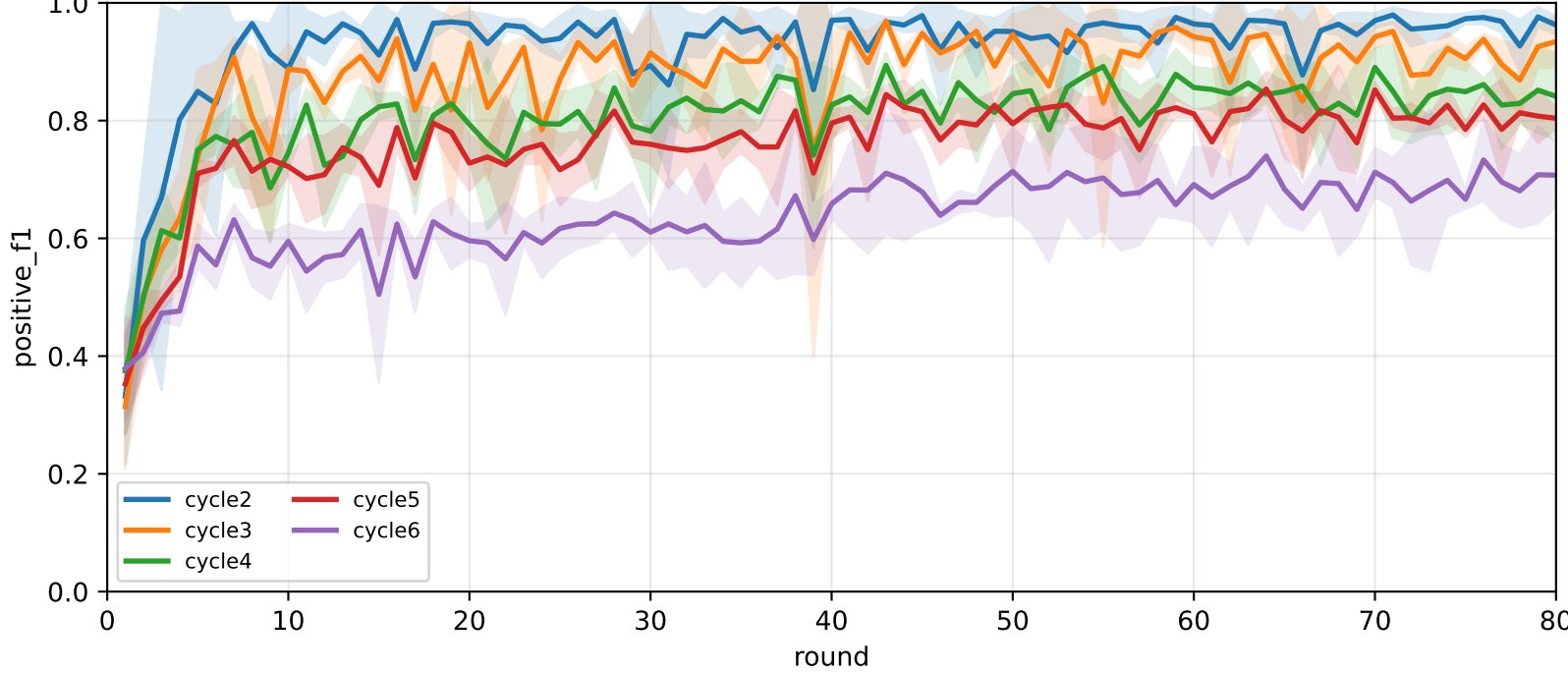
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



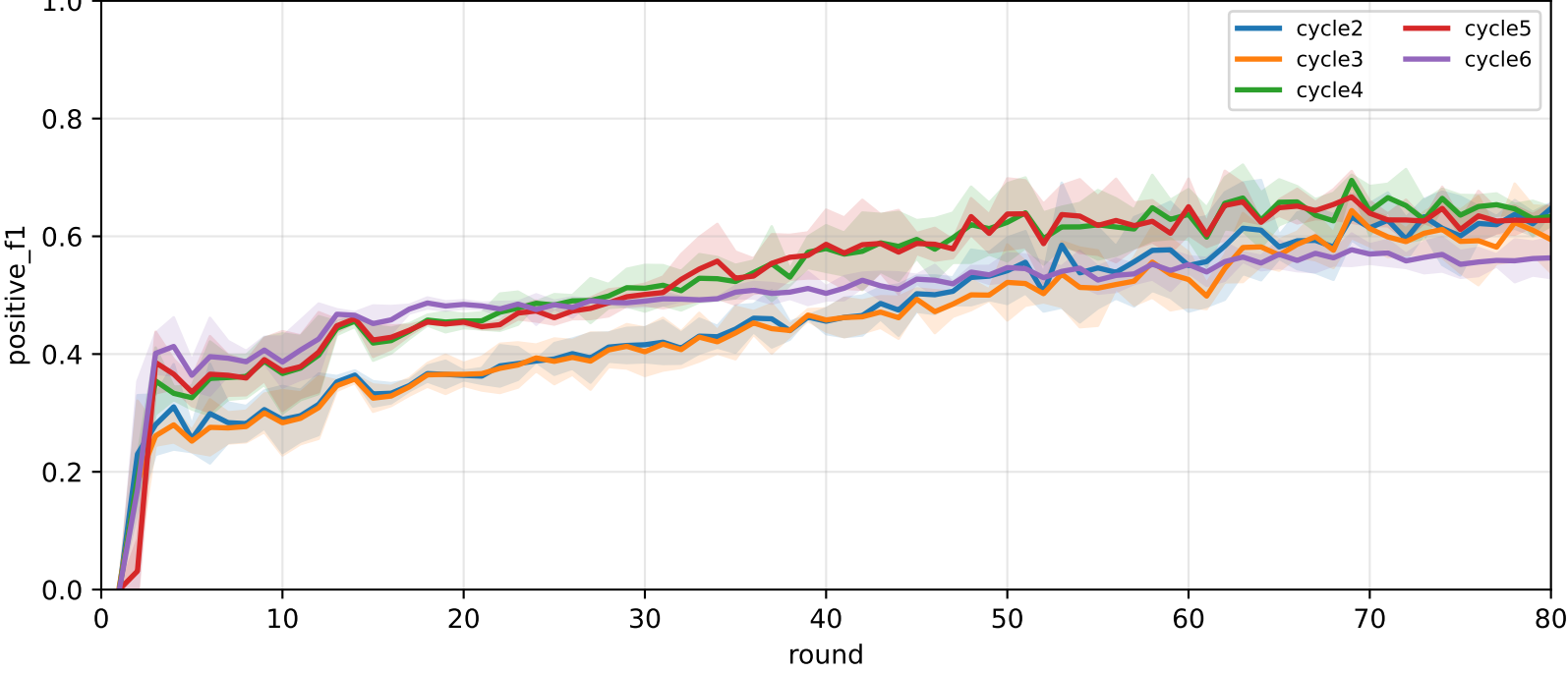
Pooled Fully Local positive F1



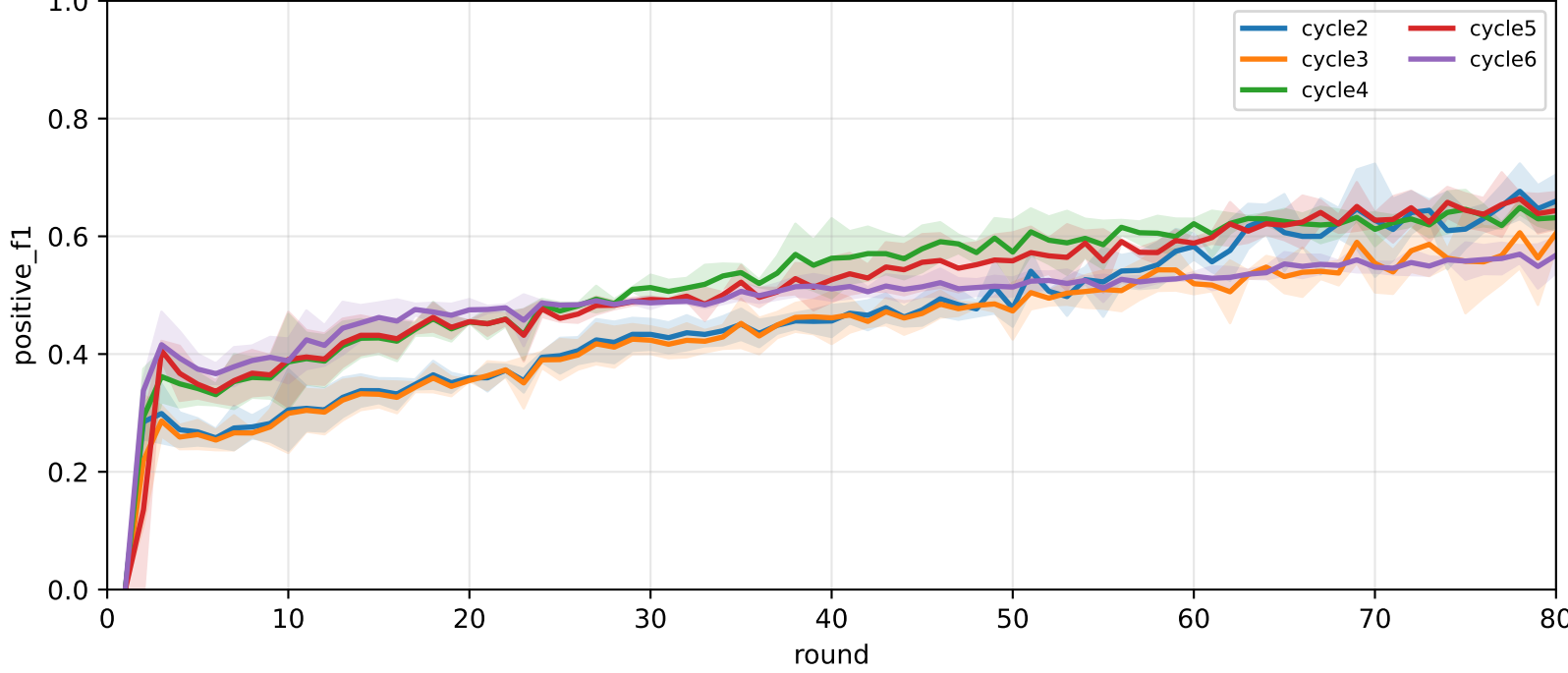
Pooled Local-Centralized positive F1



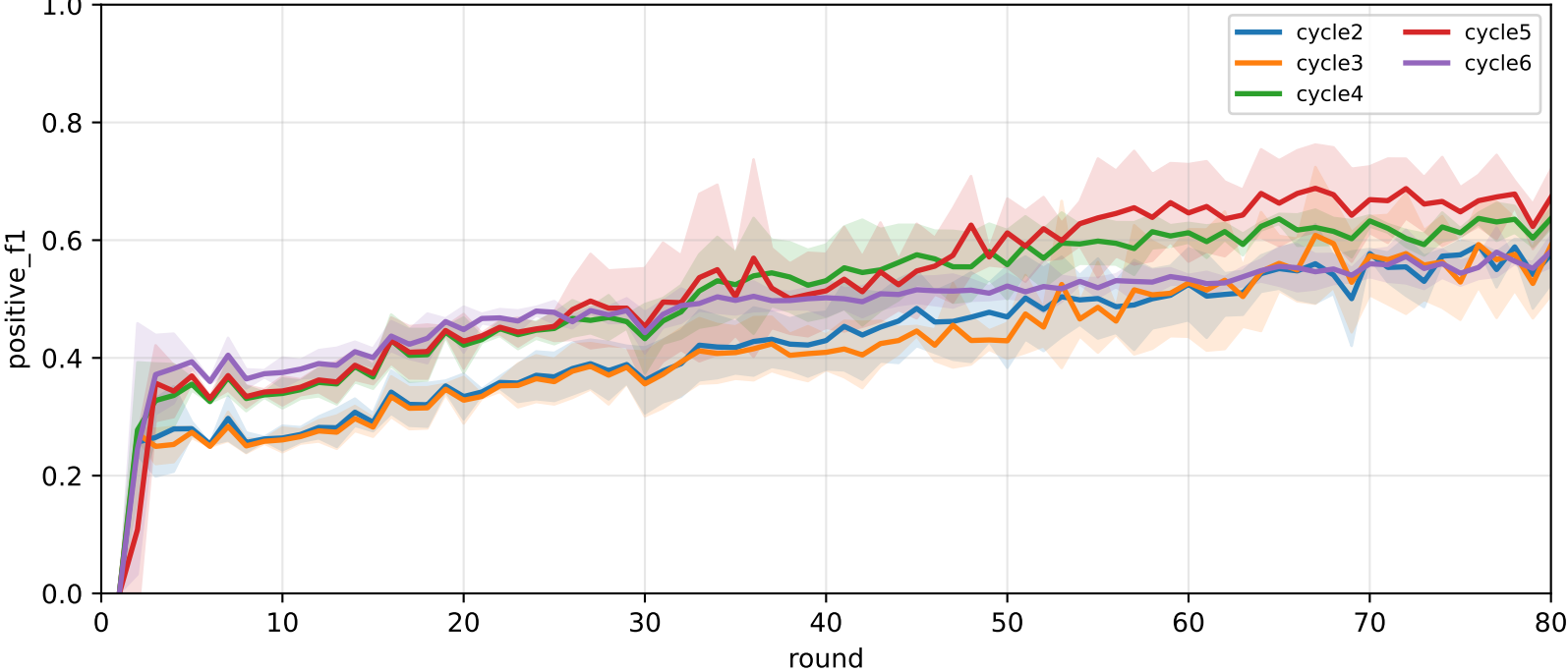
Pooled FedAvg positive F1



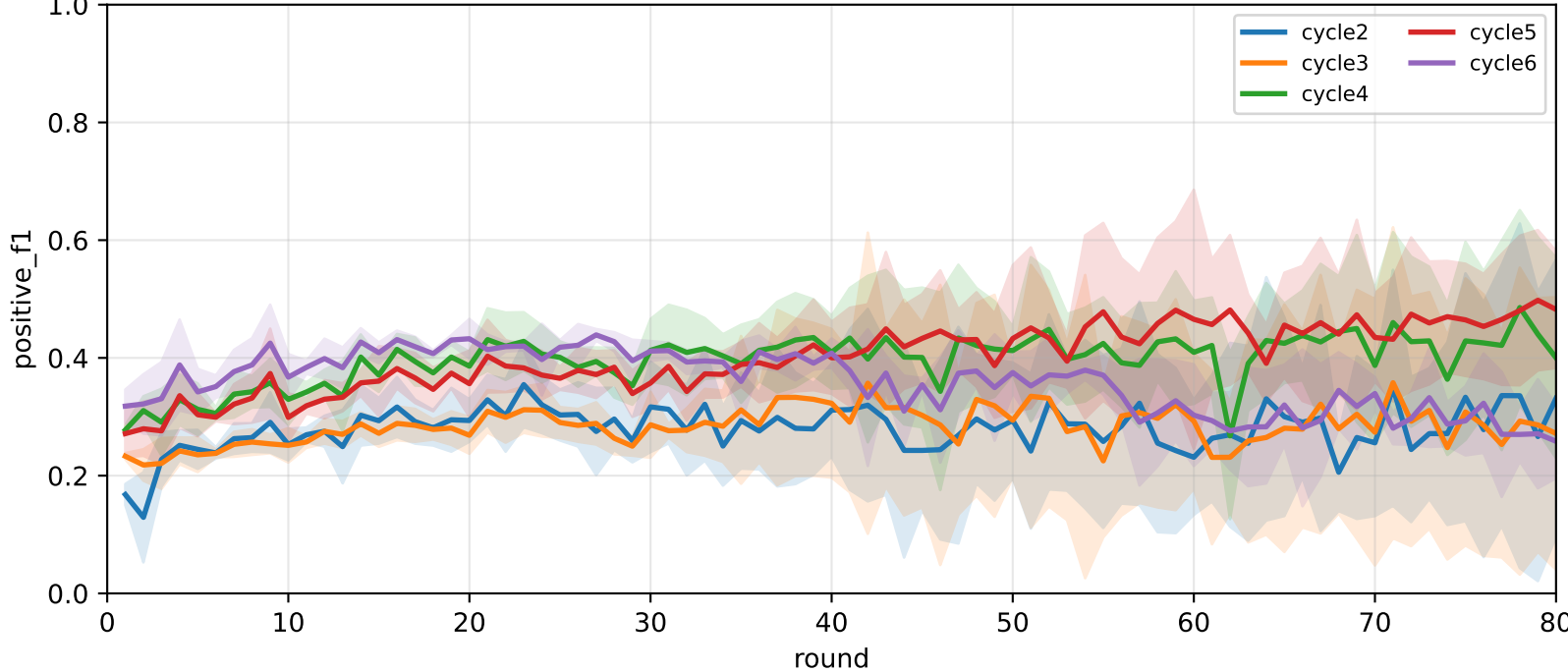
Pooled FedProx positive F1



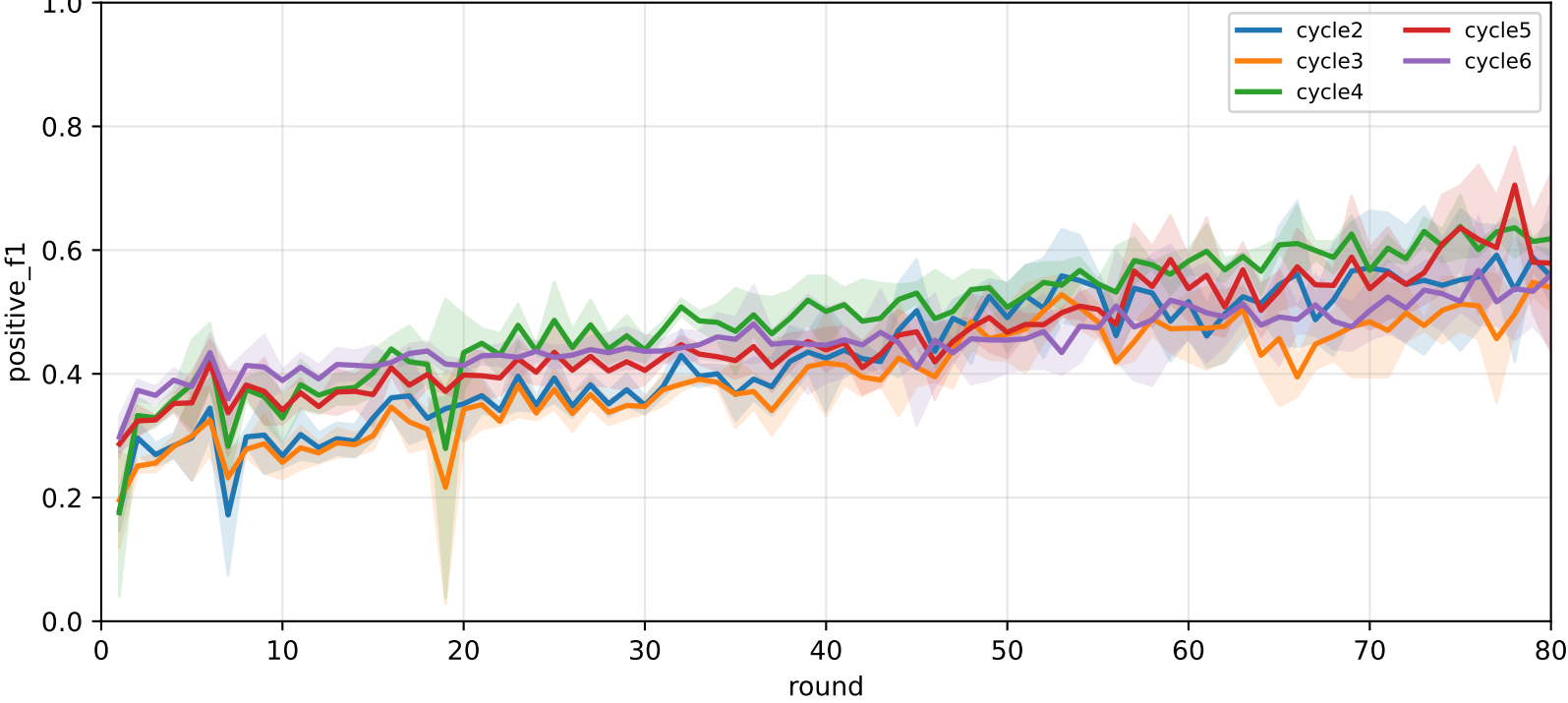
Pooled GCFL+ positive F1



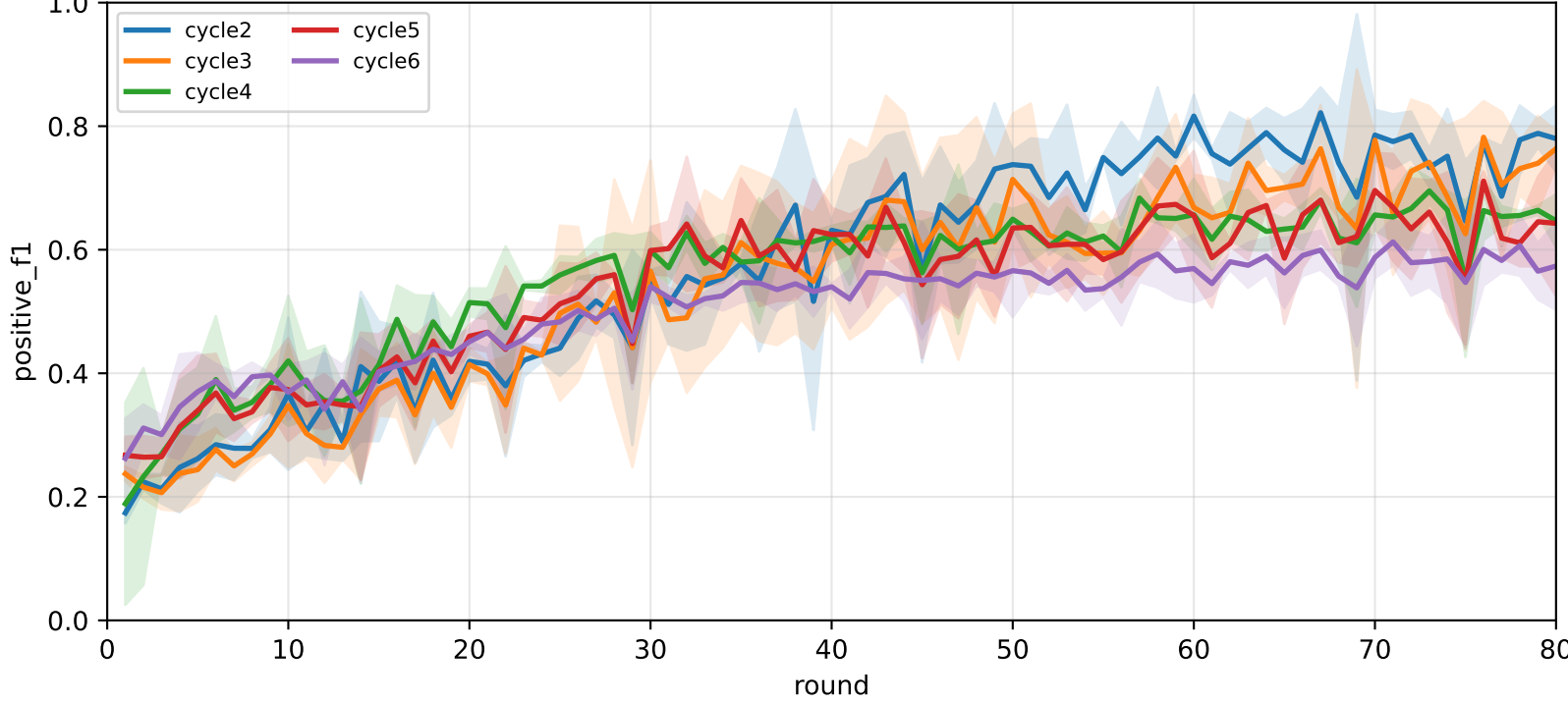
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



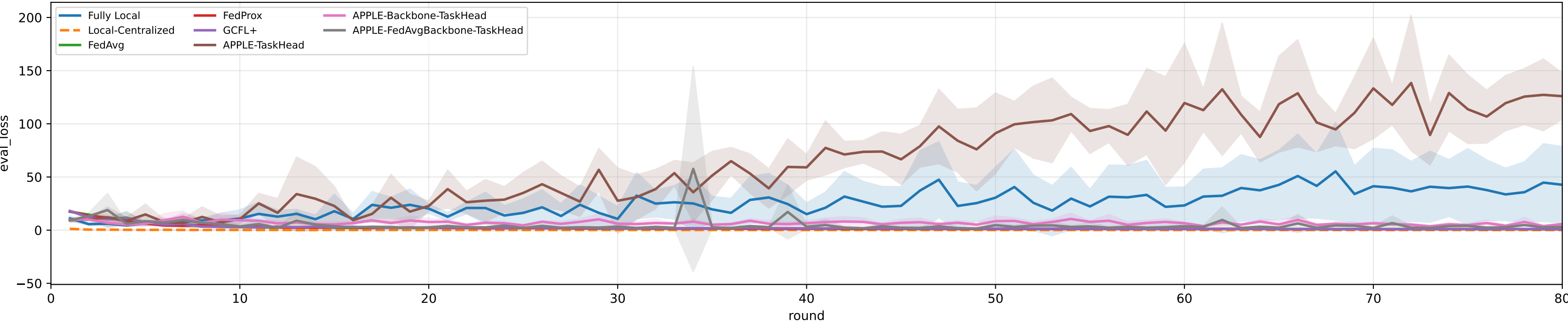
Controlled label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

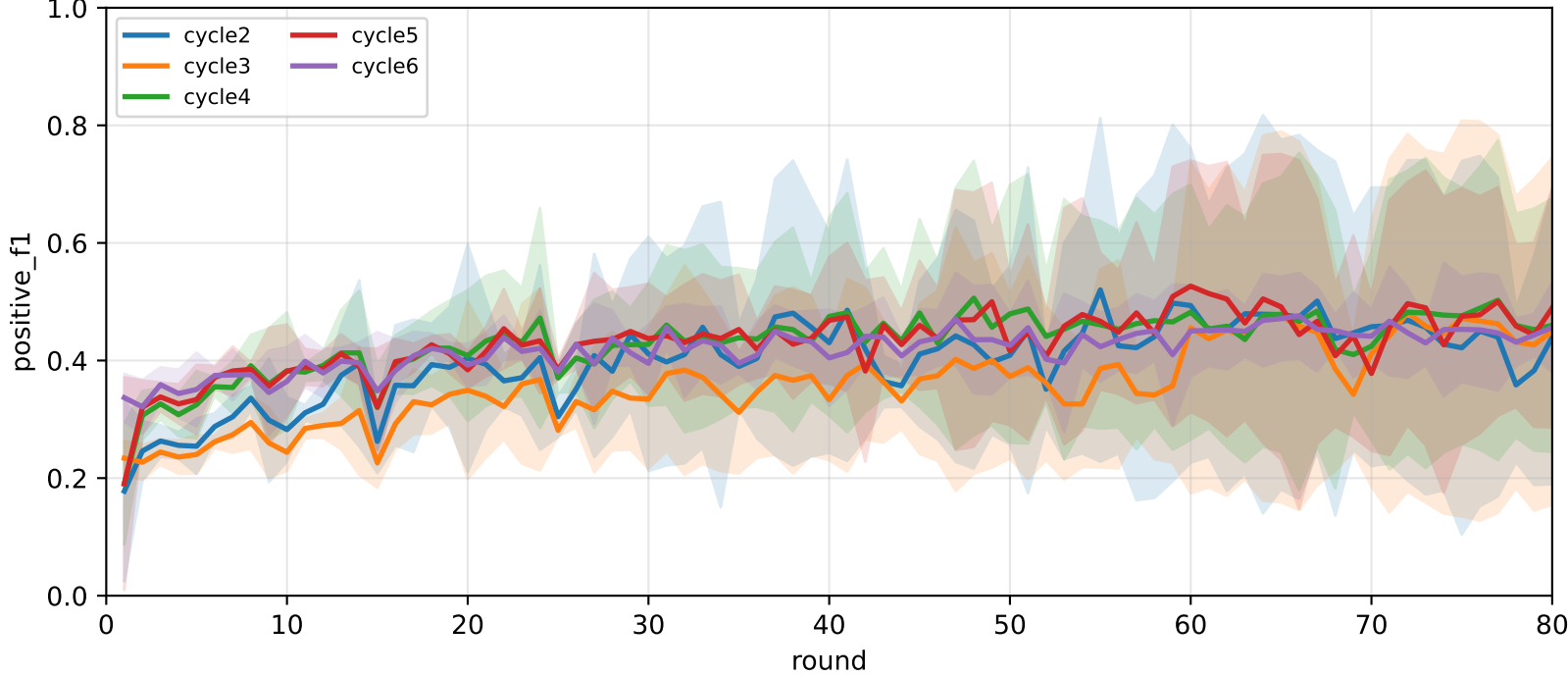
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	31.2% ± 8.8%	31.1% ± 8.9%	26.7% ± 8.1%	32.1% ± 9.5%	33.6% ± 11.2%	31.3% ± 9.2%	32.1% ± 6.8%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
FedAvg	<u>59.4% ± 2.6%</u>	<u>60.1% ± 2.5%</u>	60.8% ± 2.3%	<u>65.2% ± 6.1%</u>	<u>61.7% ± 2.0%</u>	59.8% ± 3.1%	<u>53.0% ± 3.9%</u>
FedProx	57.7% ± 1.4%	58.5% ± 1.1%	<u>62.3% ± 2.7%</u>	60.4% ± 3.8%	60.3% ± 1.7%	58.8% ± 2.5%	50.8% ± 2.7%
GCFL+	57.4% ± 2.6%	58.0% ± 2.8%	55.8% ± 1.7%	60.7% ± 6.1%	60.7% ± 1.1%	<u>61.8% ± 4.8%</u>	50.9% ± 1.2%
APPLE-TaskHead	31.9% ± 4.7%	31.8% ± 4.7%	26.1% ± 3.9%	31.9% ± 4.6%	33.6% ± 5.4%	32.8% ± 4.9%	34.7% ± 4.9%
APPLE-Backbone-TaskHead	36.6% ± 7.0%	36.5% ± 7.0%	31.3% ± 8.8%	37.2% ± 7.0%	37.7% ± 7.3%	36.6% ± 5.8%	39.5% ± 5.8%
APPLE-FedAvgBackbone-TaskHead	54.2% ± 6.7%	56.3% ± 7.6%	<u>62.3% ± 13.4%</u>	61.0% ± 17.4%	55.4% ± 4.6%	57.5% ± 7.3%	45.2% ± 3.9%

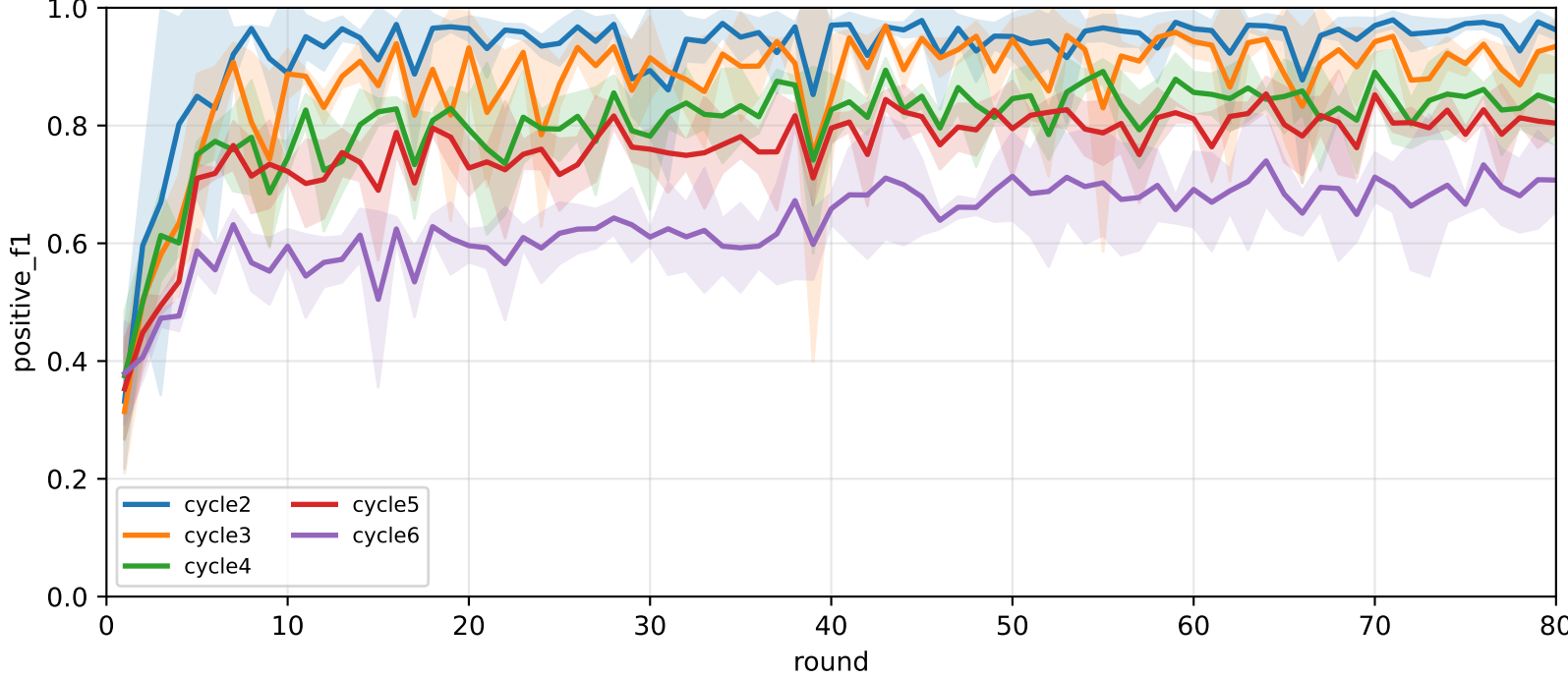
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



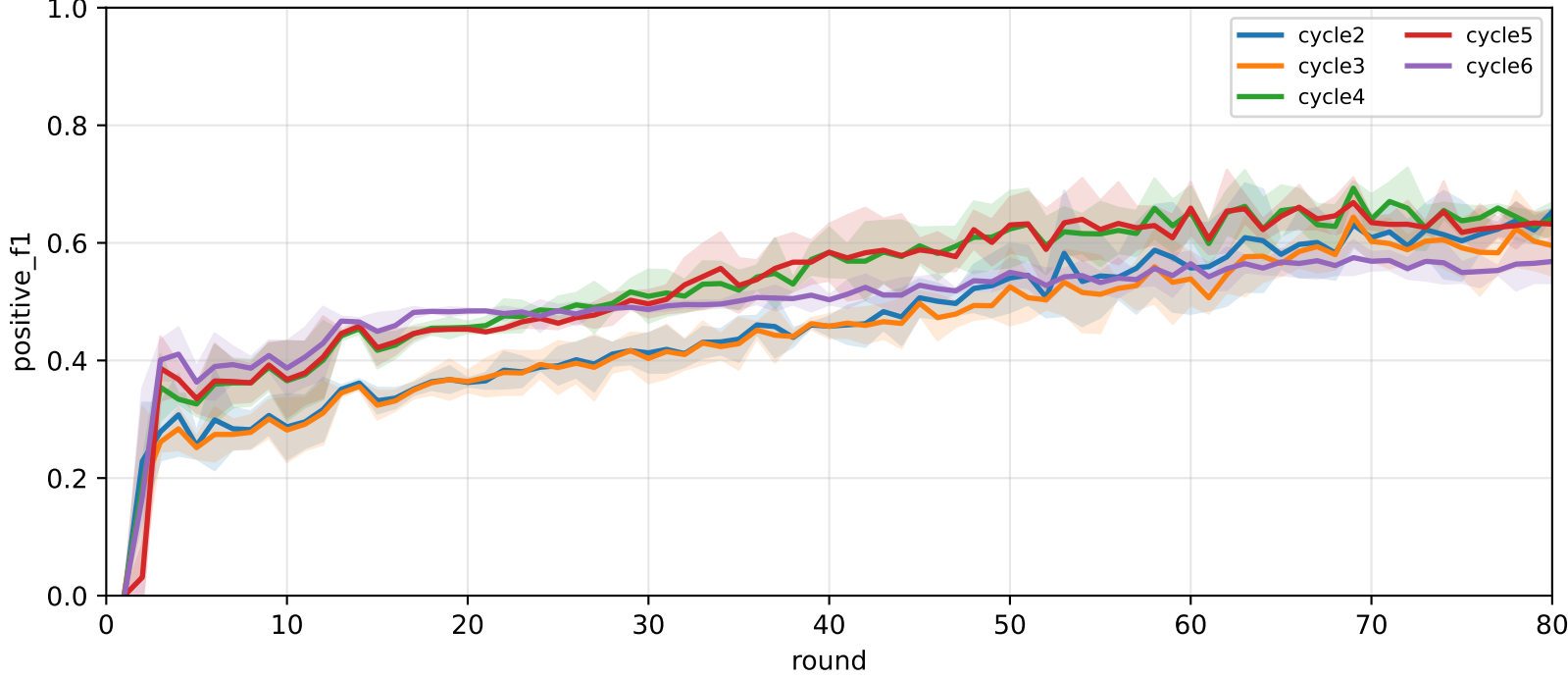
Pooled Fully Local positive F1



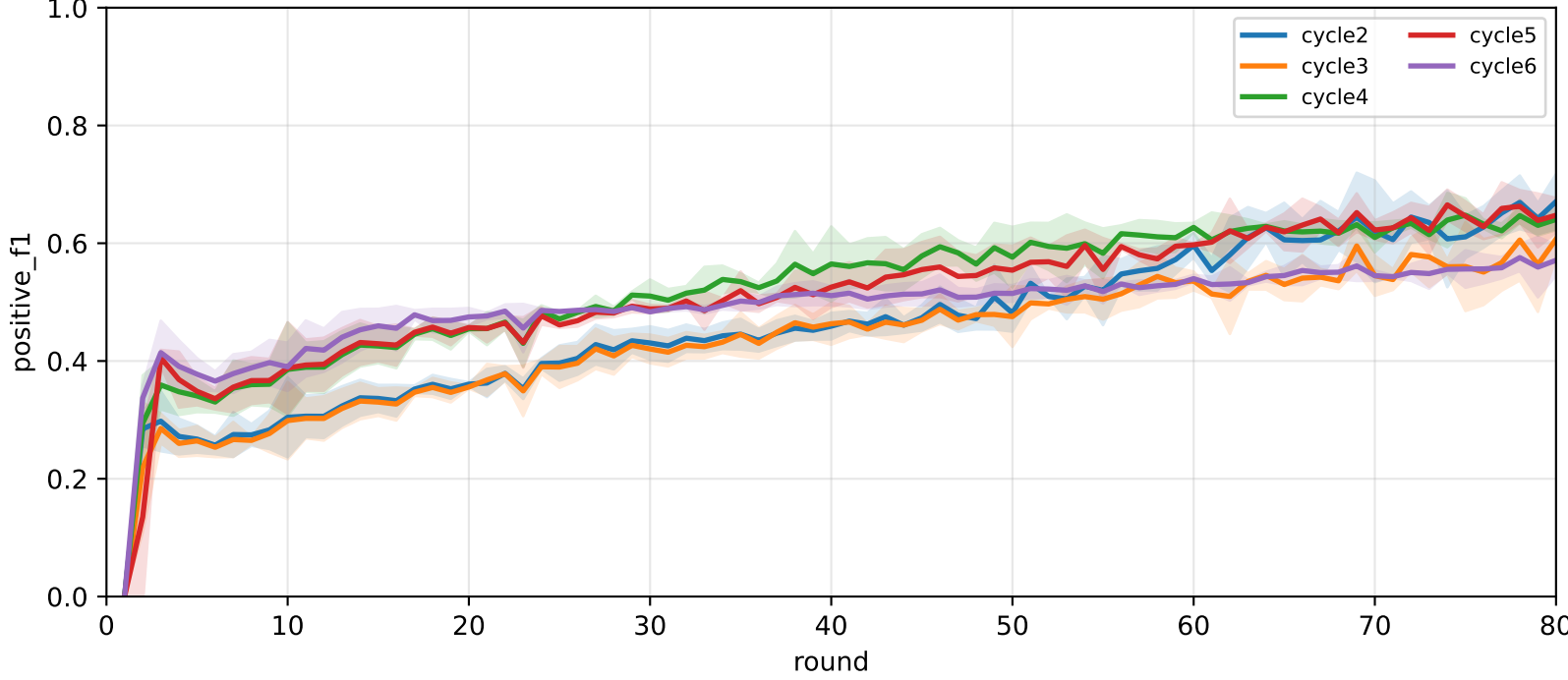
Pooled Local-Centralized positive F1



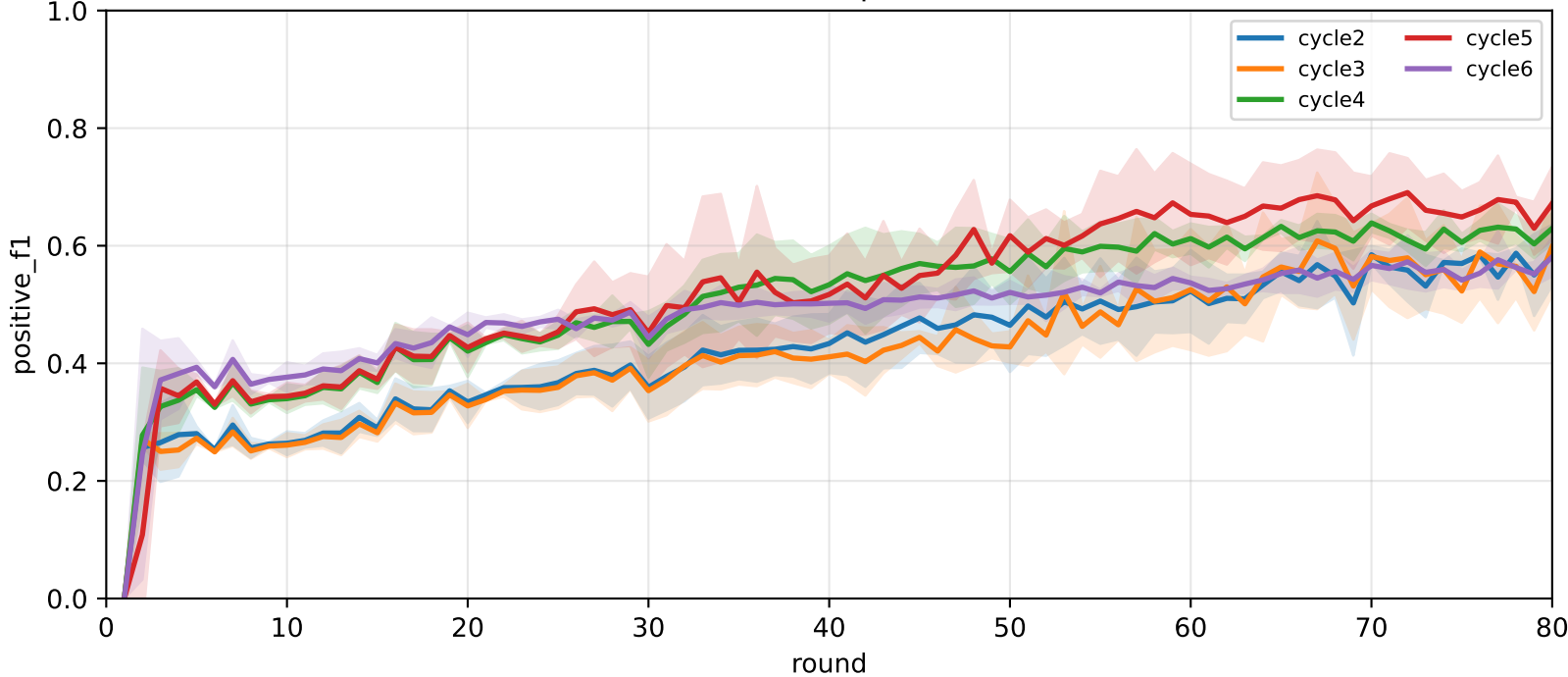
Pooled FedAvg positive F1



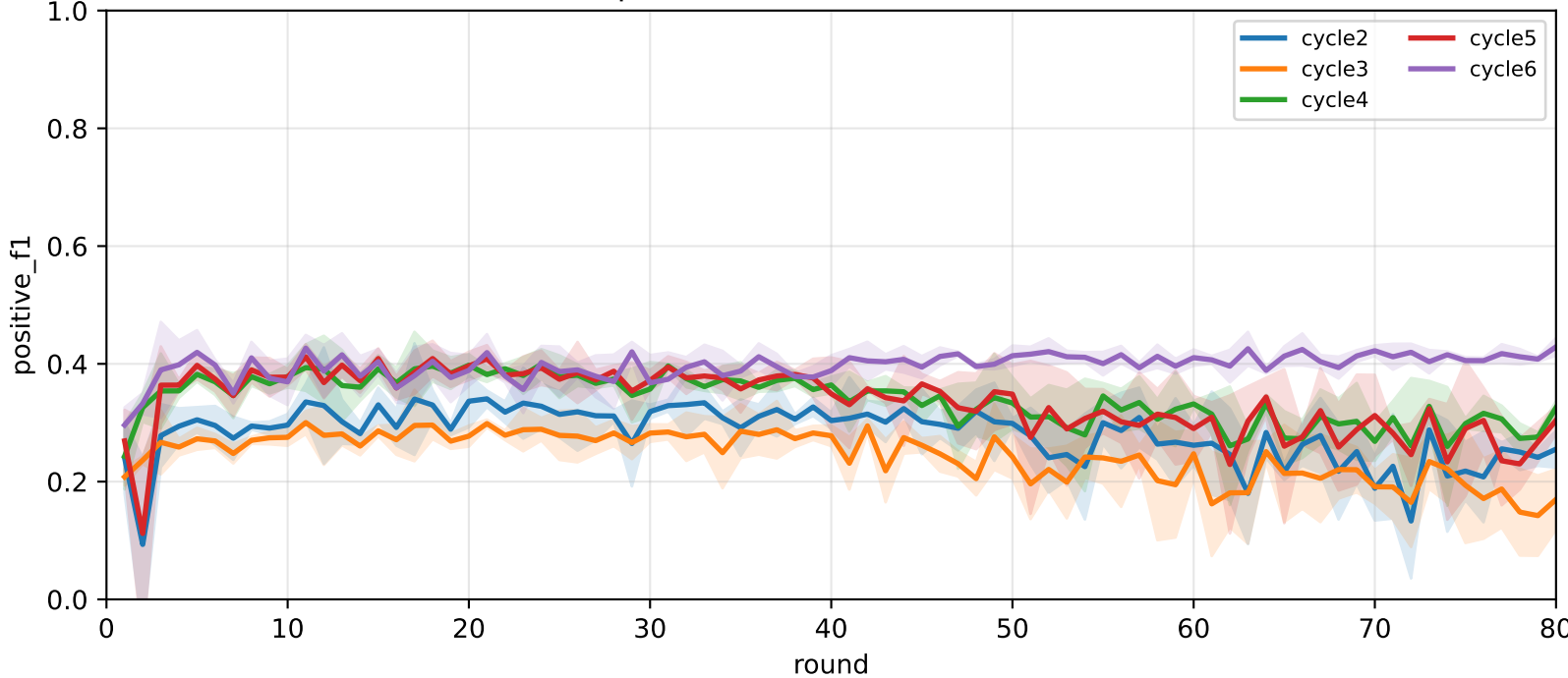
Pooled FedProx positive F1



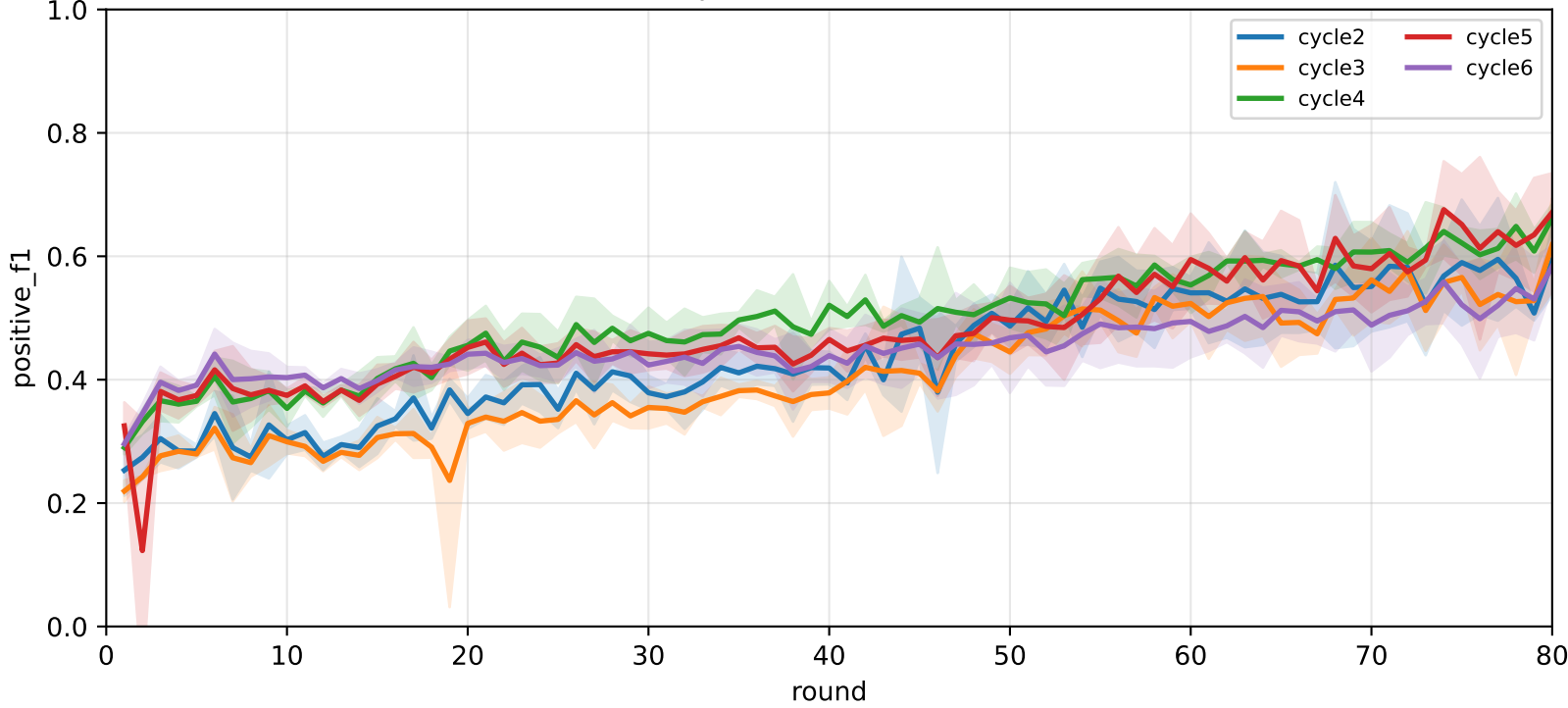
Pooled GCFL+ positive F1



Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

